Block Diagram

Company: Huaqin	
Block Title	Block Diagram
Project Doha	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 1 of 46

I2C Configuration Table

I2C	GPIO	Function	Material	I2C Slave Address(7-bit mode)
SPK_I2C_SDA/CLK	GPIO 4/GPIO 5	Smart PA	FS1862(1st)	0x34
			TAS2558(2nd)	0x4C
		Sar sensor	96T346HW (1st)	0x20
			SX9325(2nd)	0x28
		TOF	VL53L1CBV0FY/1	0x52
GSENSOR_I3C_SDA/SCL	GPIO 22/GPIO 23	A+G	LSM6DSOW (1st)	0x6A
			ICM-42605 (2nd)	0x10
SENSOR_I2C_SDA/SC	GPIO 28/GPIO 29	M sensor	AK09918(1st)	0x0C
			MMC5603NJ(2nd)	0x30
		PS	LTR-2568ALS-WA(1st)	0x23
			TBD(2nd)	TBD
		ALS	TSL2540(1st)	0x39
			MN29005(2st)	0x49

SPI Configuration Table

SPI	Function
QUD1,SE3	LCD interface
QUD1,SE1	FPS interface

I2S Configuration Table

I2S	Function
I2S	AUDIO PA interface

Table of Content

Sheet	Content	Sheet	Content
01.	Block Diagram	24.	AUDIO_MIC_RCV_SPK
02.	I2C Configuration Table	25.	AUDIO_HEADSET
03.	Table of Content	26.	SENSORS_MOTOR
04.	GPIO MAP	27.	LCD_TP
05.	SM6125 Control/UFS/SDC1/SDC2	28.	REAR_CAM/FLASH
06.	SM6125 LPDDR4	29.	FRONT_CAM/CAM_POWER
07.	SM6125 CSI/DSI	30.	SIM_SD_KEY_LINK_BATT
08.	SM6125 GPIO (Internal Codec)	31.	DTV
09.	SM6125 Power (Core)	32.	NFC
10.	SM6125 Power (Other, LP4X)	33:	PRIMARY_ANTENNA
11.	SM6125 Ground	34:	PA & PA THERMAL
12.	PM6125 CLKs/Control_GND	35:	TRX_LB
13.	PM6125 Buck Converter1	36:	TRX_MHB
14.	PM6125 Buck Converter2	37:	DIVERSITY_ANT
15.	PM6125 LDO	38:	DRX_MB_HB
16.	PM8008 and Bob	39:	DRX_LB
17.	SIM_SD	40:	SDR660_MSS
18.	PMI632 Control/Display/GPIO	41:	SDR660_PWR
19	PM660L Display	42:	SDR660_DA/PRX/DRX/FBRX
20.	eMCP	43:	QFE4101
21.	30W CHARGE	44:	WIFI_2.4G_5G & BT
22.	AOV	45:	GPS
23.	Test Points	46:	WCN3980

Company: Huaqin			
Block Title	Table of Content		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	3 of 46

GPIO MAP

SDM630 GPIO Configuration

GPIO_0	BLSP_SPI_MOSI_1	GPIO_42	NC	GPIO_84	UIM2_CLK
GPIO_1	BLSP_SPI_MISO_1	GPIO_43	NC	GPIO_85	UIM2_RESET
GPIO_2	BLSP_SPI_CS1_N_1	GPIO_44	NC	GPIO_86	**UIM2_PRESENT**
GPIO_3	BLSP_SPI_CLK_1	GPIO_45	NC	GPIO_87	**UIM1_DATA**
GPIO_4	BLSP_UART_TX_2	GPIO_46	CAM0_RST_N	GPIO_88	UIM1_CLK
GPIO_5	BLSP_UART_RX_2	GPIO_47	CAM1_RST_N	GPIO_89	UIM1_RESET
GPIO_6	BLSP_I2C_SDA_2	GPIO_48	CAM2_RST_N	GPIO_90	UIM1_PRESENT
GPIO_7	BLSP_I2C_SCL_2	GPIO_49	NC	GPIO_91	UIM_BATT_ALARM
GPIO_8	BLSP_SPI_MOSI_3	GPIO_50	CAM_AF_VDD_EN	GPIO_92	**NC**
GPIO_9	BLSP_SPI_MISO_3	GPIO_51	CAM_AVDD_EN	GPIO_93	**NC**
GPIO_10	BLSP_SPI_CS_N_3	GPIO_52	NC	GPIO_94	**NC**
GPIO_11	BLSP_SPI_CLK_3	GPIO_53	LCD0_RESET_N	GPIO_95	**NC**
GPIO_12	NC	GPIO_54	SD_CARD_DET_N	GPIO_96	**WDOG_DISABLE**
GPIO_13	KEYPAD_LED_PWM	GPIO_55	DP_EN_N	GPIO_97	**NC**
GPIO_14	BLSP_I2C_SDA_4	GPIO_56	USBC_ORIENTATION	GPIO_98	**NC**
GPIO_15	BLSP_I2C_SCL_4	GPIO_57	FORCED_USB_BOOT	GPIO_99	**QLINK_REQUEST**
GPIO_16	BLSP_UART_TX_5	GPIO_58	USB_PHY_PS	GPIO_100	**QLINK_ENABLE**
GPIO_17	BLSP_UART_RX_5	GPIO_59	MDP_VSYNC_P	GPIO_101	**RFFE1_DATA - SDR660 debug**
GPIO_18	BLSP_UART_CTS_N_5	GPIO_60	NC	GPIO_102	**RFFE1_CLK - SDR660 debug**
GPIO_19	BLSP_UART_RFR_N_5	GPIO_61	NC	GPIO_103	**RFFE2_DATA - (41)QAT3533**
GPIO_20	FP_SUB_RESET	GPIO_62	NC	GPIO_104	**RFFE2_CLK - (41)QAT3533**
GPIO_21	SMB_STAT	GPIO_63	NC	GPIO_105	**NC**
GPIO_22	BLSP_I2C_SDA_6	GPIO_64	NC	GPIO_106	**NC**
GPIO_23	BLSP_I2C_SCL_6	GPIO_65	NC	GPIO_107	**RFFE4_DATA-(45)QSW8574,(48)RF1656SR**
GPIO_24	CDC_SWR_CLK	GPIO_66	TS_RESET_N	GPIO_108	**RFFE4_CLK -(45)QSW8574,(48)RF1656SR**
GPIO_25	CDC_SWR_DATA	GPIO_67	TS_INT_N	GPIO_109	**RFFE5_DATA-(52)QAT3514, (52)QAT3550,(47) QAT3522**
GPIO_26	WSA_SPKR_SD_N_1	GPIO_68	ACC_GYRO_DATA_AVA_INT_N	GPIO_110	**RFFE5_CLK-(52)QAT3514, (52)QAT3550,(47)QAT3522**
GPIO_27	NC	GPIO_69	ACC_GYRO_MOT_INT_N	GPIO_111	**RFFE6_DATA - (46)QET4101 ch0, (41)QPA4361**
GPIO_28	NFC_IRQ	GPIO_70	MAG_INT_N	GPIO_112	**RFFE6_CLK - (46)QET4101 ch0, (41)QPA4361**
GPIO_29	NFC_EN	GPIO_71	ALSP_INT_N	GPIO_113	**NC**
GPIO_30	NFC_DWL_REQ	GPIO_72	FP_INT_N_1	LPI_GPIO_0	LPI_SPI_1_CS1_N
GPIO_31	NFC_ESE_PWR_REQ	GPIO_73	NC	LPI_GPIO_1	LPI_PWR_EN
GPIO_32	CAM_MCLK0	GPIO_74	NC	LPI_GPIO_2	LPI_I2C_3_SDA
GPIO_33	CAM_MCLK1	GPIO_75	NC	LPI_GPIO_3	LPI_I2C_3_SCL
GPIO_34	CAM_MCLK2	GPIO_76	NC	LPI_GPIO_4	LPI_SPI_2_CS_N
GPIO_35	NC	GPIO_77	AUDIO_USBC_EN2_N	LPI_GPIO_5	LPI_SPI_2_CLK
GPIO_36	CCI_I2C_SDA0	GPIO_78	WMSS_RESETN	LPI_GPIO_6	LPI_SPI_2_MOSI
GPIO_37	CCI_I2C_SCL0	GPIO_79	**PWM_CNTRL**	LPI_GPIO_7	LPI_SPI_2_MISO
GPIO_38	CCI_I2C_SDA1	GPIO_80	**AUDIO_USBC_EN1**	LPI_GPIO_8	LPI_SPI_1_CS_N
GPIO_39	CCI_I2C_SCL1	GPIO_81	MSS_LTE_COXM_TXD	LPI_GPIO_9	LPI_SPI_1_CLK
GPIO_40	NC	GPIO_82	MSS_LTE_COXM_RXD	LPI_GPIO_10	LPI_SPI_1_MOSI
GPIO_41	FL_STROBE_TRIG	GPIO_83	UIM2_DATA	LPI_GPIO_11	LPI_SPI_1_MISO

LPI_GPIO_12	LPI_UART_1_TX
LPI_GPIO_13	LPI_UART_1_RX
LPI_GPIO_14	NC
LPI_GPIO_15	NC
LPI_GPIO_16	NC
LPI_GPIO_17	NC
LPI_GPIO_18	CDC_PDM_CLK
LPI_GPIO_19	CDC_PDM_SYNC
LPI_GPIO_20	CDC_PDM_TX
LPI_GPIO_21	CDC_PDM_RX0
LPI_GPIO_22	CDC_PDM_RX0_COMP
LPI_GPIO_23	CDC_PDM_RX1
LPI_GPIO_24	CDC_PDM_RX1_COMP
LPI_GPIO_25	CDC_PDM_RX2
LPI_GPIO_26	CDC_DMIC_CLK1
LPI_GPIO_27	CDC_DMIC_DATA1
LPI_GPIO_28	CDC_DMIC_CLK2
LPI_GPIO_29	CDC_DMIC_DATA2
LPI_GPIO_30	LPI_QCA_SB_CLK
LPI_GPIO_31	LPI_QCA_SB_DATA0

PM660 GPIO Configuration

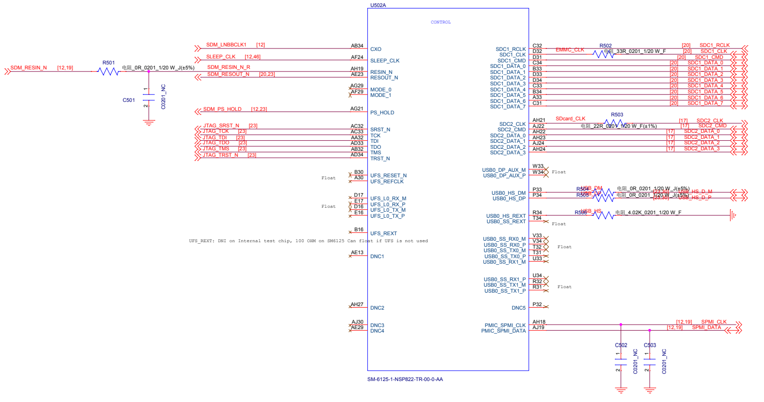
GPIO_1	OPTION1	GPIO_8	SLB
GPIO_2	DIV_CLK2	GPIO_9	uUSB_TYPEC
GPIO_3	NC	GPIO_10	WCSS_VCTRL
GPIO_4	NFC_CLK_REQ	GPIO_11	HOMEKEY_FP_PM_INT
GPIO_5	WLAN_SW_CTRL	GPIO_12	WIPWR_MODE
GPIO_6	SLP_CLK	GPIO_13	PM_A_GPIO_13
GPIO_7	UIM_BATT_ALARM		

PM660A GPIO Configuration

GPIO_1	OPTION2	GPIO_8	SD_CARD_DET_N
GPIO_2	LPI_PWR_EN	GPIO_9	WCSS_VCTRL
GPIO_3	FRONT_CAM_DVDD_EN	GPIO_10	SLB
GPIO_4	REAR_CAM_DVDD_EN	GPIO_11	LP4x_CTRL
GPIO_5	NC	GPIO_12	LP4x_MODE
GPIO_6	FP_VDD_EN		
GPIO_7	KEY_VOL_UP_N		

SM6125 Control/UFS/SDC1/SDC2

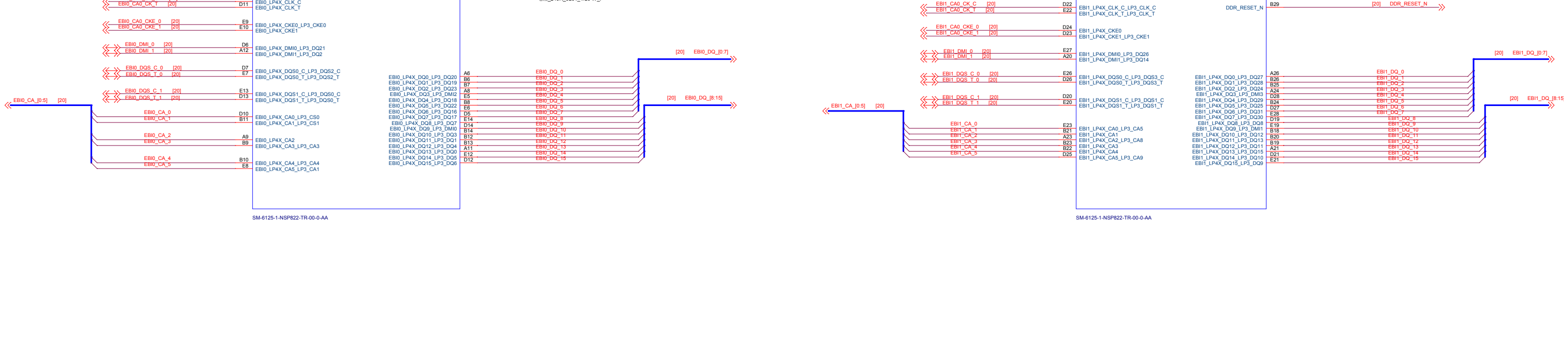
80-PJ285-41 Rev. A



Company: Huaqin			
Block Title	SDM630 Control/UFS/SDC1/SDC2		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	5 of 46

SM6125 LPDDR4

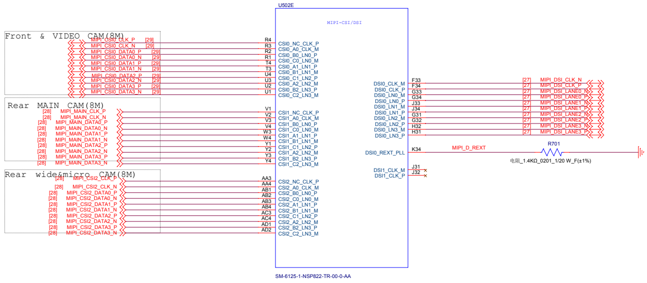
80-PJ285-41 Rev. A

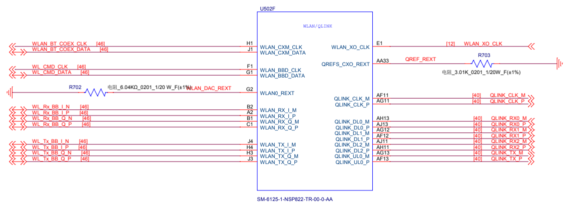


Company: Huaqin				
Block Title	SDM630 LPDDR4			
Project	Teller	Rev 1.0		
Date:	Saturday, July 27, 2019	Sheet	6 of	46

SM6125 CSI/DSI

REF 701-799

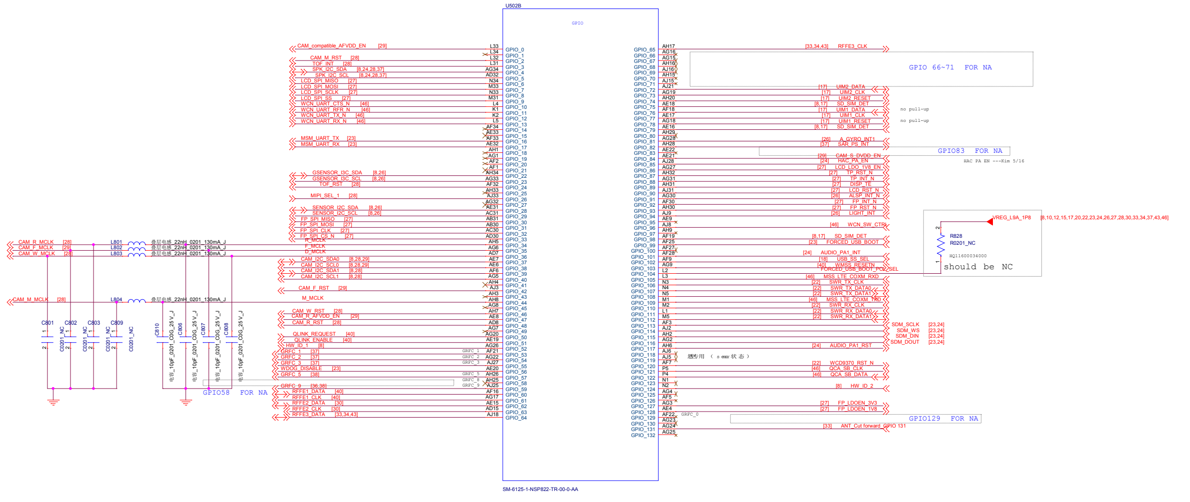




Company: Huaqin			
Block Title	SDM630 CSI/DSI		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	7 of 46

SM6125 GPIO (Internal Codec)

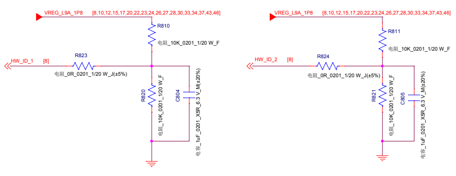
REF 801-899



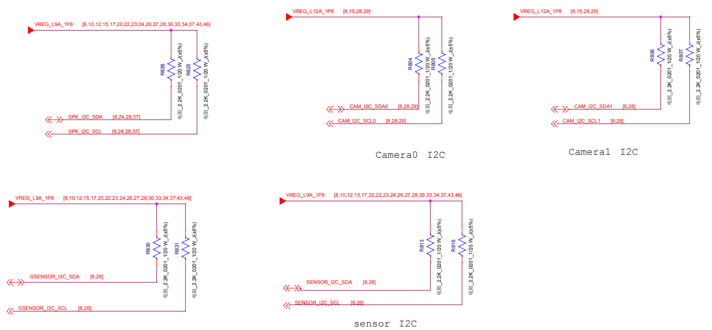
GPIO_56	BOOT_CONFIG(0)	WDOG_DISABLE
GPIO_87	BOOT_CONFIG(1)	FASTBOOT_SEL(0)
GPIO_32	BOOT_CONFIG(2)	FASTBOOT_SEL(1)
GPIO_118	BOOT_CONFIG(3)	FASTBOOT_SEL(2)
GPIO_99	FORCED_USB_BOOT	
GPIO_104	FORCED_USB_BOOT_POL_SEL	

BOOT OPTIONS	BOOT CONFIG BITS 3	2	1
0: Default(eMMC->SD->USB(EDL))	0	0	0
1: SD->eMMC->USB(EDL)	0	0	1
2: SD->USB(EDL)	0	1	0
3: USB(EDL)	0	1	1
4: UFS->SD->USB(EDL)	1	0	0
5: SD->UFS->USB(EDL)	1	0	1
6: UFS->USB(EDL)	1	1	0
7: UFS_PWM_MODE->USB(EDL)	1	1	1

Hardware ID



I2C pull up



Camera0 I2C

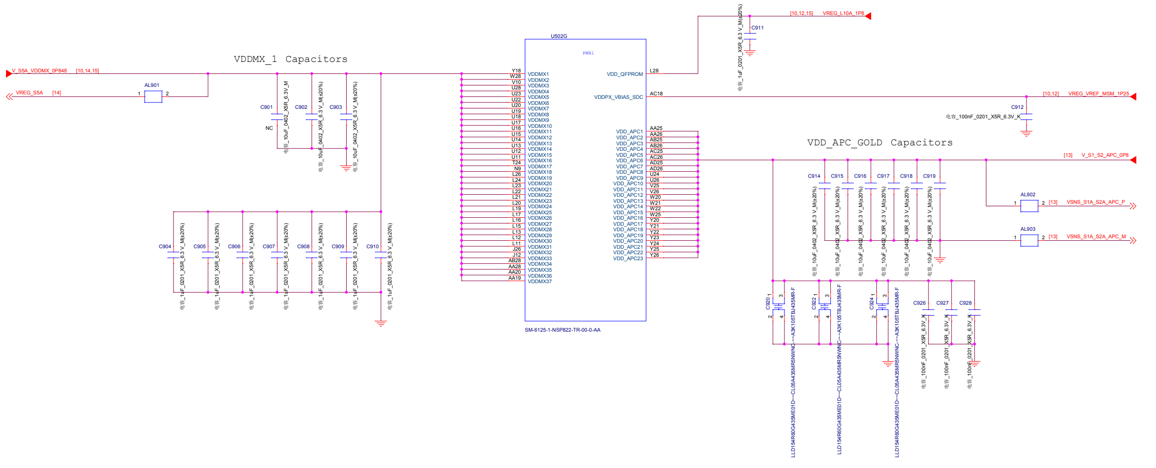
Camera1 I2C

sensor I2C

Company: Huaqin			
Block Title	SDM630 GPIO (Internal Codec)		
Project	Teller	Rev 1.0	
Date:	Saturday, July 27, 2019	Sheet 8	of 46

SM6125 Power (Core)

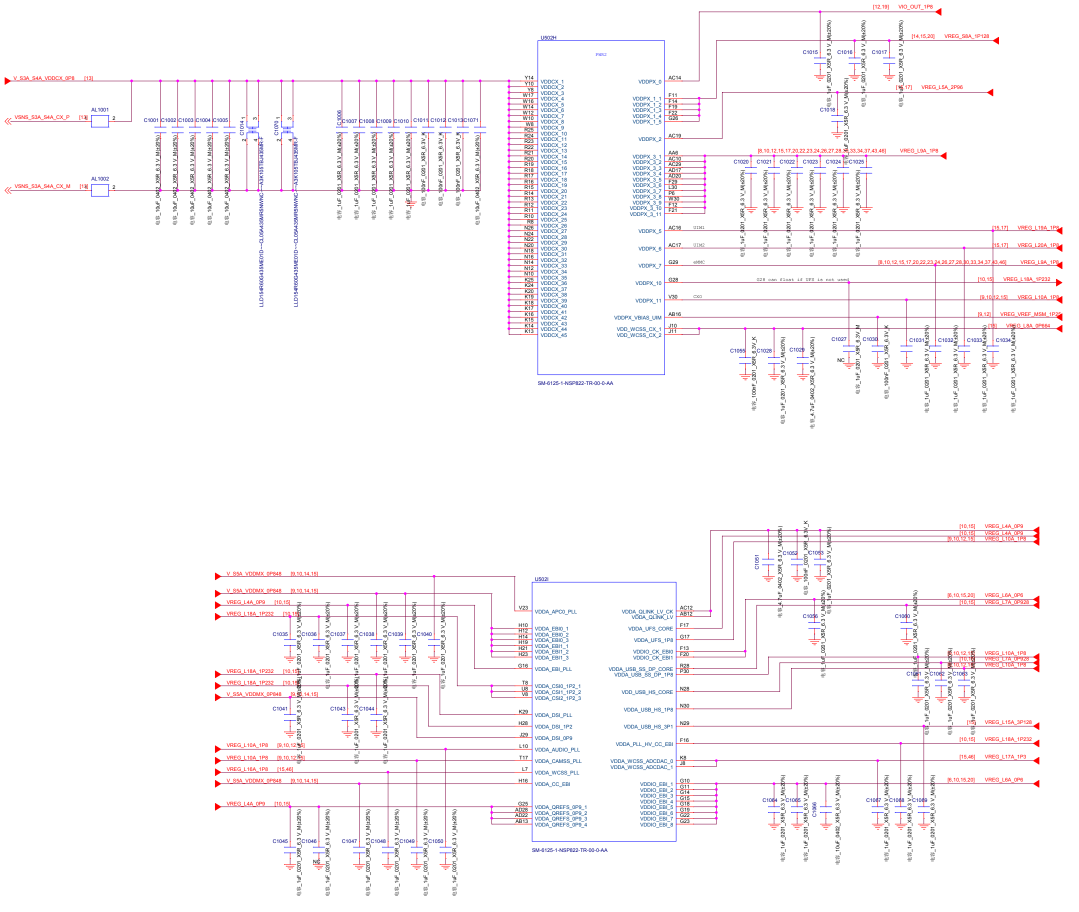
REF 901-999



Company: Huaqin	
Block Title	SDM630 Power (Core)
Project Teller	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 9 of 46

SM6125 Power (Other, LP4X)

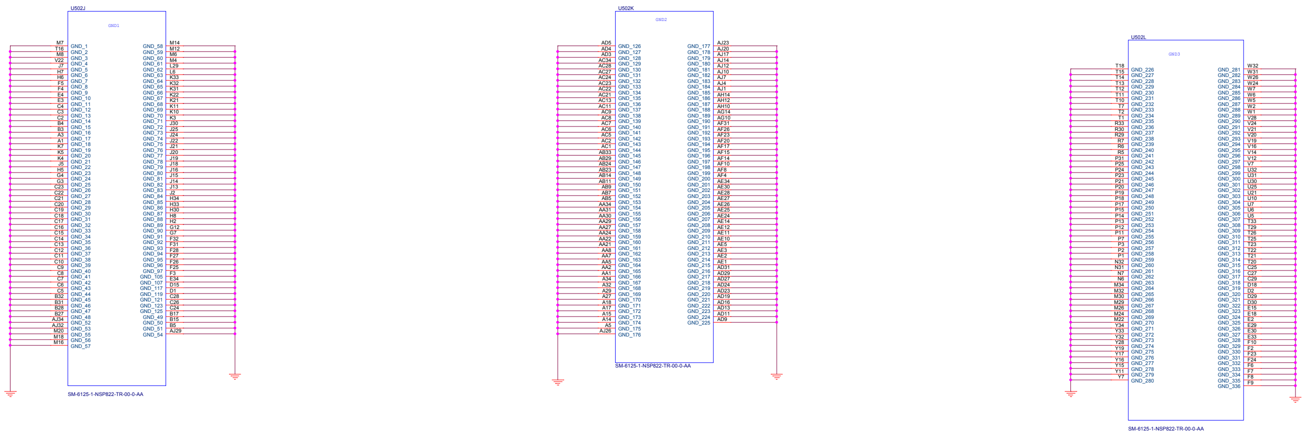
REF 1001-1099



Company: Huaqin			
Block Title	SDM630 Power (Other, LP4X)		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	10 of 46

SM6125 Ground

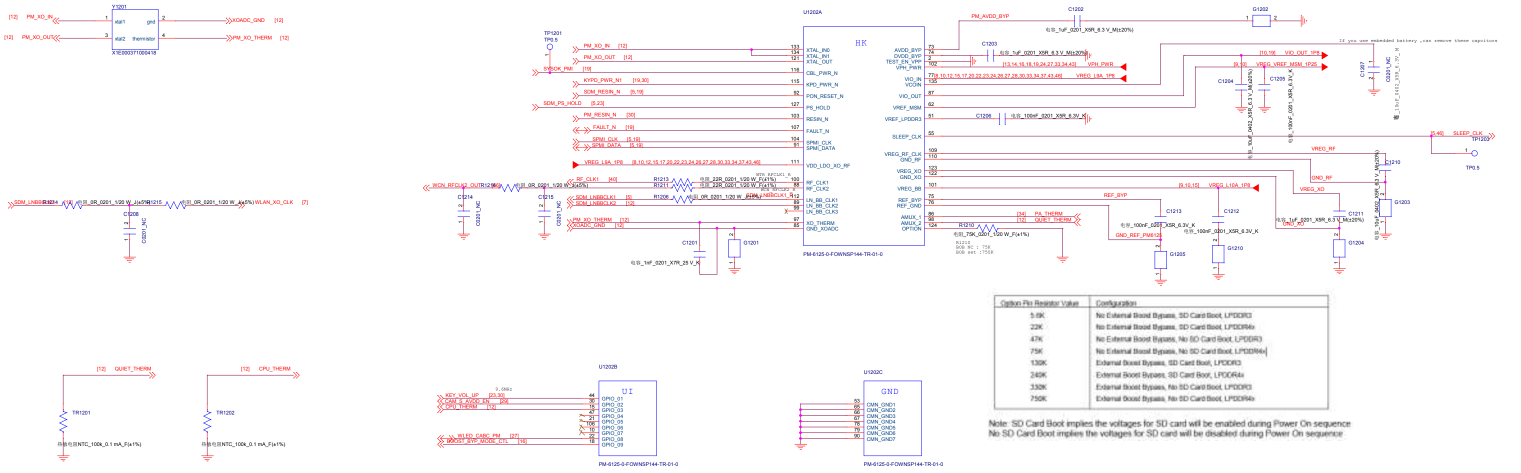
REF 1101-1199



Company: Huaqin	
Block Title	SDM630 Ground
Project	Teller
Rev	1.0
Date:	Saturday, July 27, 2019
Sheet	11 of 46

PM6125 CLKs/Control_GND

REF 1201-1299



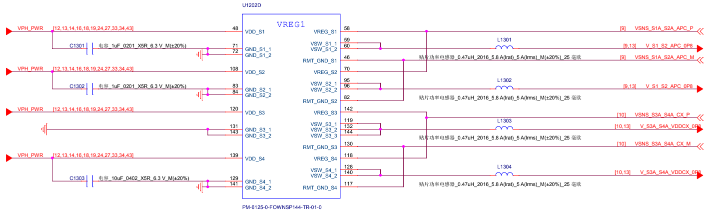
Option Pin Resistor Value	Configuration
5.6K	No External Boost Bypass, SD Card Boot, LPDDR3
22K	No External Boost Bypass, SD Card Boot, LPDDR4x
47K	No External Boost Bypass, No SD Card Boot, LPDDR3
75K	No External Boost Bypass, No SD Card Boot, LPDDR4x
130K	External Boost Bypass, SD Card Boot, LPDDR3
240K	External Boost Bypass, SD Card Boot, LPDDR4x
330K	External Boost Bypass, No SD Card Boot, LPDDR3
750K	External Boost Bypass, No SD Card Boot, LPDDR4x

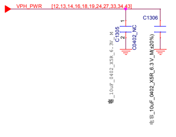
Note: SD Card Boot implies the voltages for SD card will be enabled during Power On sequence
No SD Card Boot implies the voltages for SD card will be disabled during Power On sequence

Company: Huaqin	
Block Title	PM660 CLKs/Control_GND
Project Teller	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 12 of 46

PM6125 Buck Converter

REF 1301-1399





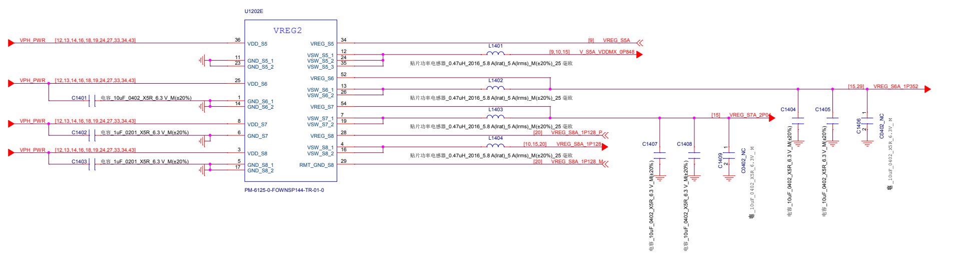
Nicobar Power Grid v1.31 10/31/2018

S1A-2A:	FTS510	0.8V default	8000mA	Ipeak 8110.00mA	APC
S3A-4A:	FTS510	0.8V default	8000mA	Ipeak 8480mA	CX, MODEM
S5A:	HFS510	0.912V default	4000mA	Ipeak 3620.00mA	EBI, MX, nLDO
S6A:	HFS510	1.352V default	4000mA	Ipeak 1560mA	LDO
S7A:	HFS510	2.04V default	2500mA	Ipeak 2350mA	LDO
S8A:	FTS510	1.128V default	4000mA	Ipeak 1790mA	LDO, LPDDR4x

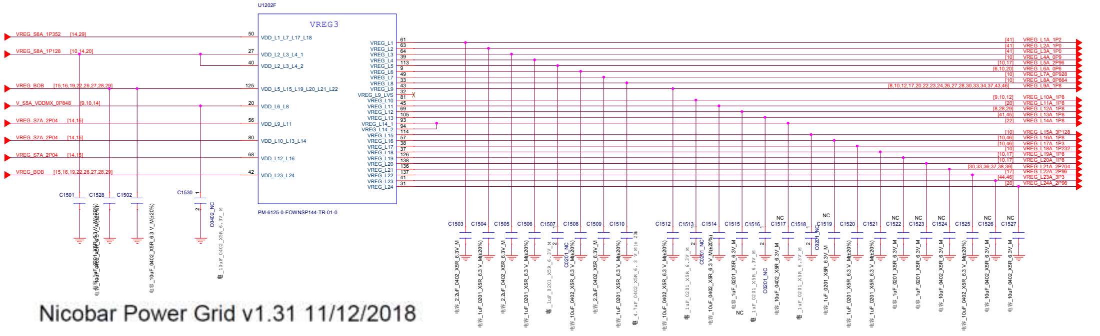
Company: Huaqin	
Block Title	PM660 Buck Converter
Project Teller	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 13 of 46

PM6125 LDO Circuits

REF 1401-1499



Company: Huaqin			
Block Title	PM660 LDO Circuits		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	14 of 46



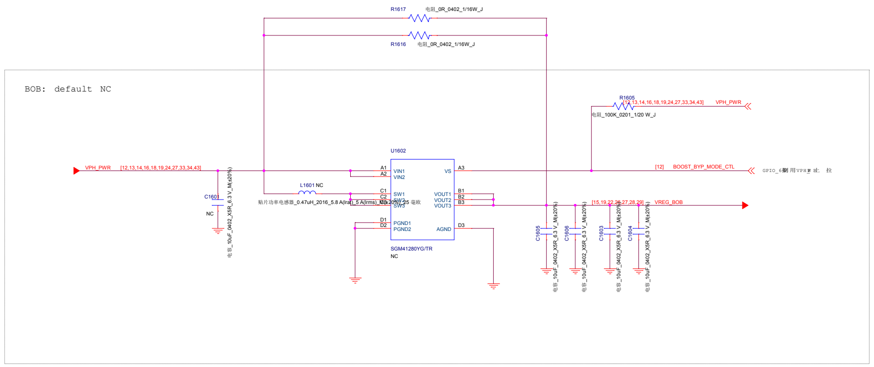
Nicobar Power Grid v1.31 11/12/2018

LDO Type Source V_default (est) Target Pin#
L1 N600 S6A 1.2V (625) QLN,SDR 1,2 61
L2 N300 S8A 1V (320) SDR VDD 1.0V 63
L3 N600 S8A 1V (370) SDR VDD 1.0V 64
L4 N300 S8A 0.928V (126) QLINK, EBI, MIPI 39
L5 MVP50 VPH 2.96V (22) PX 113
L6 N1200 S5A 0.624V (1040) VDDQ, EBI 9
L7 N600 S6A 0.928V (54.2) USB 49
L8 N300 S5A 0.664V (81) CX 33
L9 LVP600 S7A 1.8V (754) Sensors, WCD, TMD, PMI_MSM, QET 43
L10 LVP150 S7A 1.8V (100) QFPROM, USB,UFS, PX 81
L11 LVP600 S7A 1.8V (925) EMMC/VCCQ 45
L12 LVP300 S7A 1.8V (121) OV, USB_REDRIVER 69
L13 LVP300 S7A 1.8V (204) QPM 1.8V, BBRX, GPS 105
L14 LVP600 S7A 1.8V (650) WCD Buck 94,93
L15 MVP150 VPH 3.128V (5.28) USB 3.1V, DP PHY 114
L16 LVP150 S7A 1.8V (92.5) WCSS, WCN_XO 93
L17 N300 S6A 1.304V (246) WCN, WCSS ADX DAC 38
L18 N300 S6A 1.232V (133) MIPI, VDDPX 37
L19 MVP150 VPH/BB 1.8V (60.2) Memory, SN100, PX 126
L20 MVP150 VPH/BB 1.8V (60.2) Memory, SN100, PX 138
L21 MVP600 VPH/BB 2.704V (0.68) QAT, DFE, RTC,QSW 136
L22 MVP600 VPH/BB 2.96V (800) MMC 137
L23 MVP600 VPH/BB 3.304V (591) WCN 3.3V 41
L24 MVP600 VPH/BB 2.96V (1170) EMMC 31

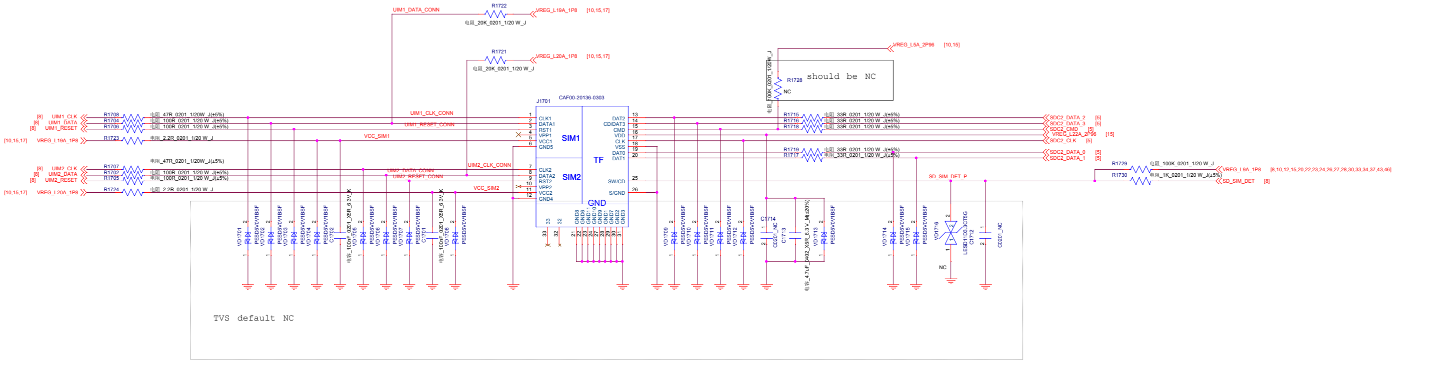
LDO L5/L9/L10/L11/L12/L13/L14/L15/L16/L19/L20/L21/L22/L23/L24 is the Pseudo-capless LDO, so can dni CAP in BOM

PSEUDO CAPLESS LDOs
P-type are psuedo-capless, cap can be at load
For CAPLESS LDOs: If decaps on the load side do not
add up to LDO spec, then install the cap close to the PMIC

Company: Huaqin			
Block Title	PM660 Charger_FG		
Project	Teller	Rev	1.0
Date:	Monday, July 29, 2019	Sheet	15 of 46

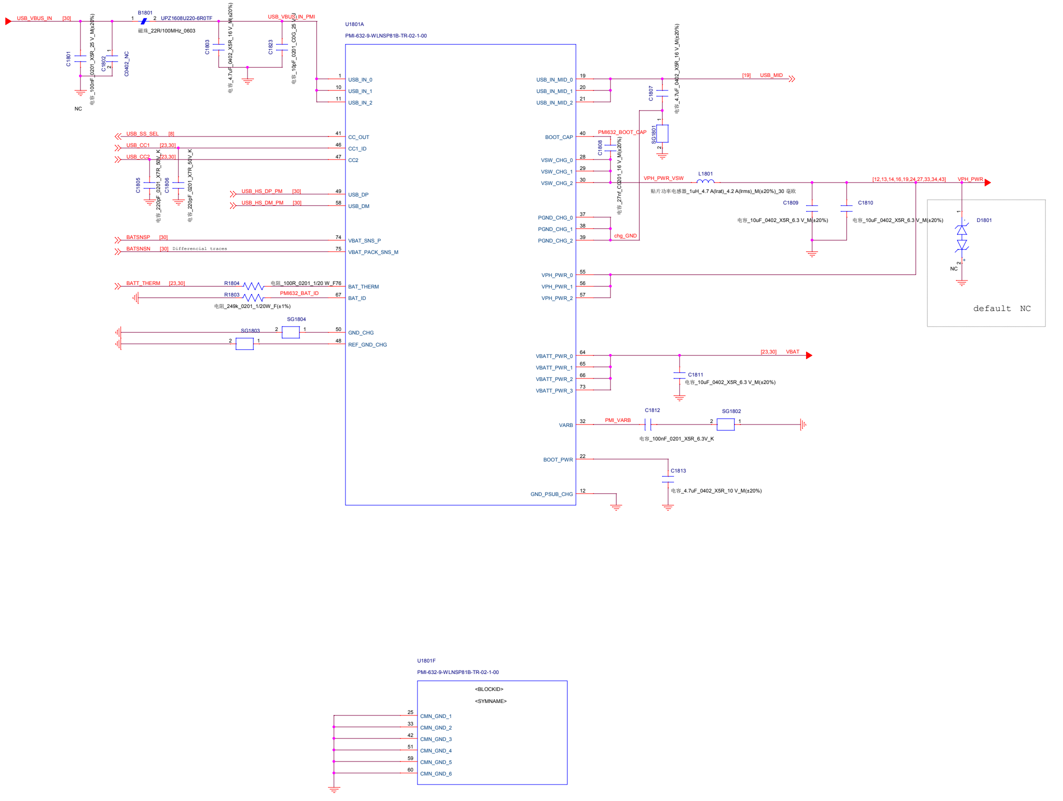


Company: Huaqin	
Block Title	PM660L Control and Bob_Flash
Project Teller	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 16 of 46



Company: Huaqin			
Block Title	PM660L Buck and LDO		
Project	Teller	Rev	1.0
Date	Saturday, July 27, 2019	Sheet	17 of 46

PMI PIN	connection on device without Wipower	connection on device with Wipower
CHG_OK	Pull up to VDD_CAP with 51kohm	Stark PRU CHG_OK
DVI2_EN	Pull down to GND with 0ohm	Stark PRU DIV2_EN
GPIO2	No Connect For parallel charging: Connects to SMB EN pin. Do NOT pull up to VDD_CAP	10Kohm pull down to gnd For parallel charging: Connects to SMB EN pin. Do NOT pull up to VDD_CAP

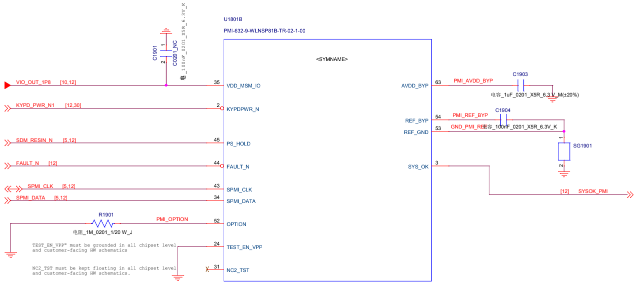


PMI632 Charger

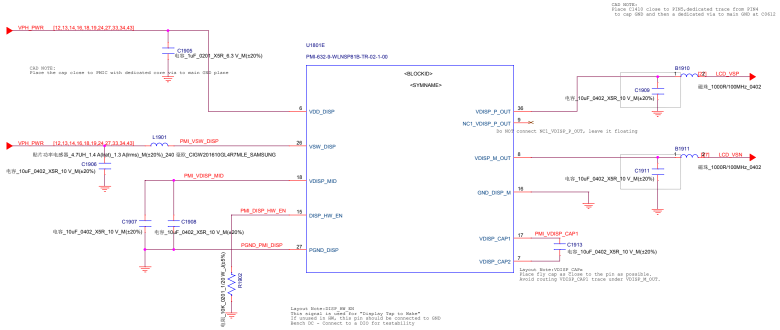
Company: Huaqin	
Block Title	PM660L CODEC
Project Teller	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 18 of 46

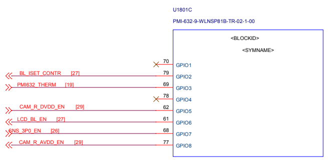
PMI632 Display

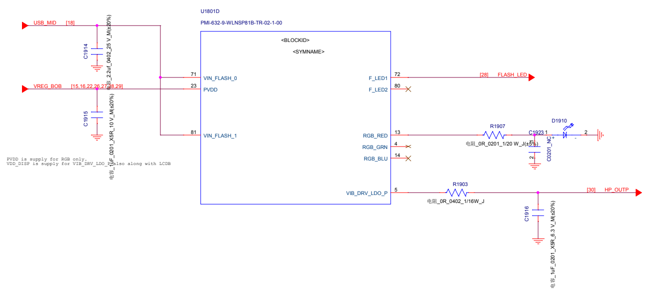
REF 80-PG427-41 Rev. B

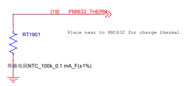


OPTION PIN	Configuration	PD Resistance
1	SPMI, uUSB	1K
2	I2C, uUSB	22K
7	I2C, TypeC	510K
	SPMI, TypeC	1M





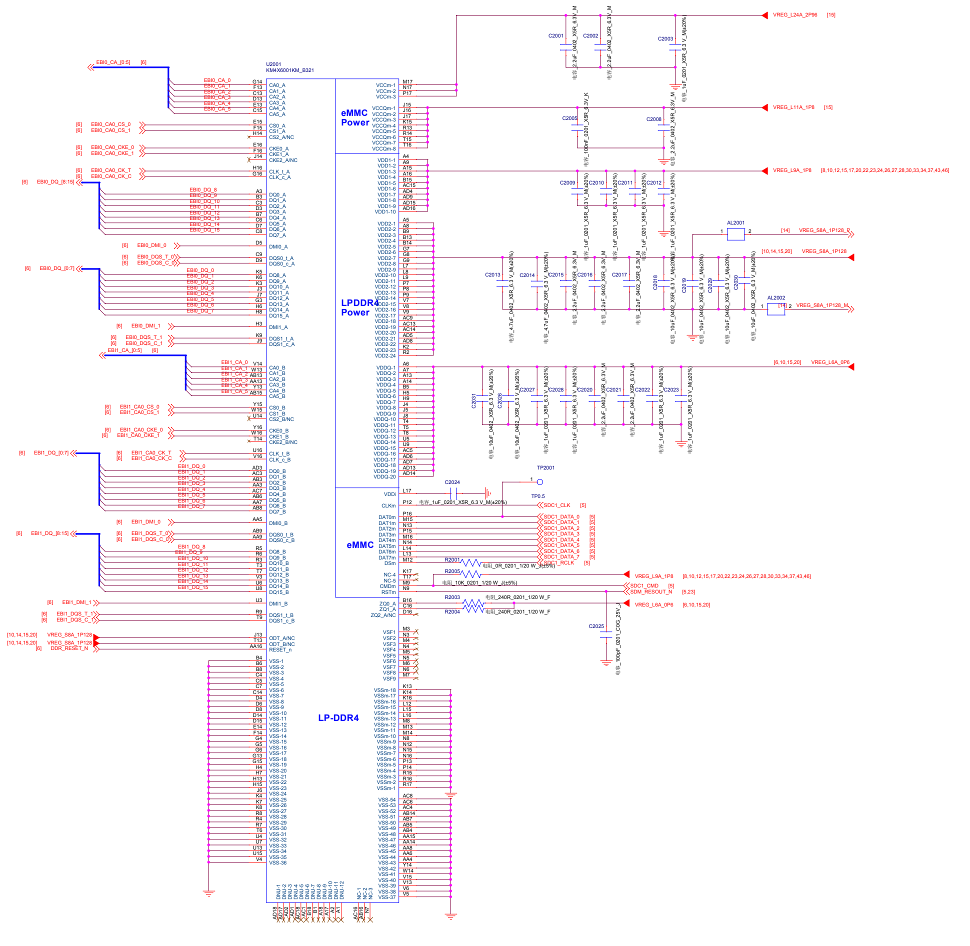




Company: Huaqin	
Block Title	PM660L Display
Project: Teller	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 19 of 46

eMCP

REF 2001-2099



Company:	Huaqin
Block Title	eMCP
Project	Teller
Rev	1.0
Date:	Saturday, July 27, 2019
Sheet	20 of 46

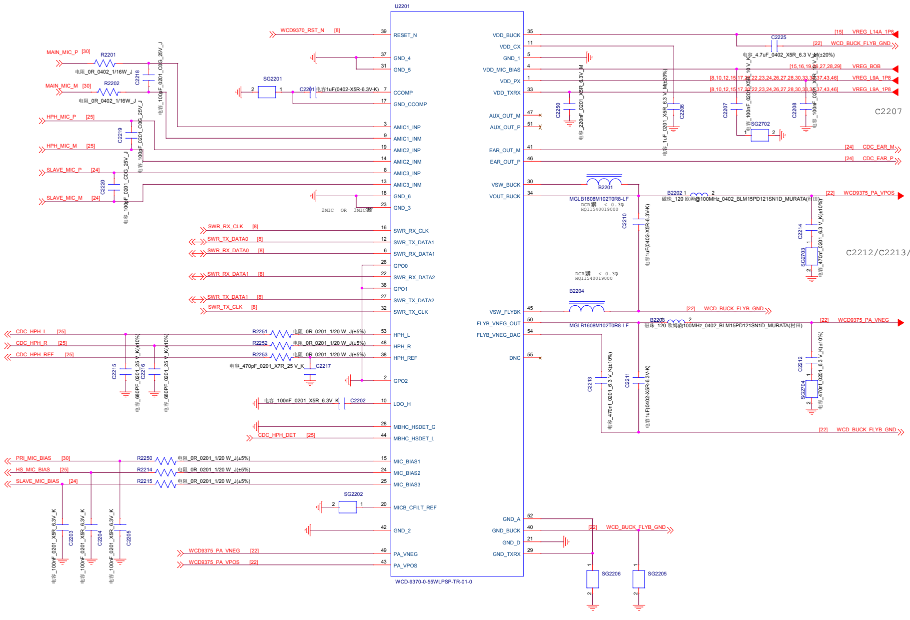
24W CHARGE

REF 2101-2199

Company: Huaqin			
Block Title	30W CHARGE		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	21 of 46

CODEC

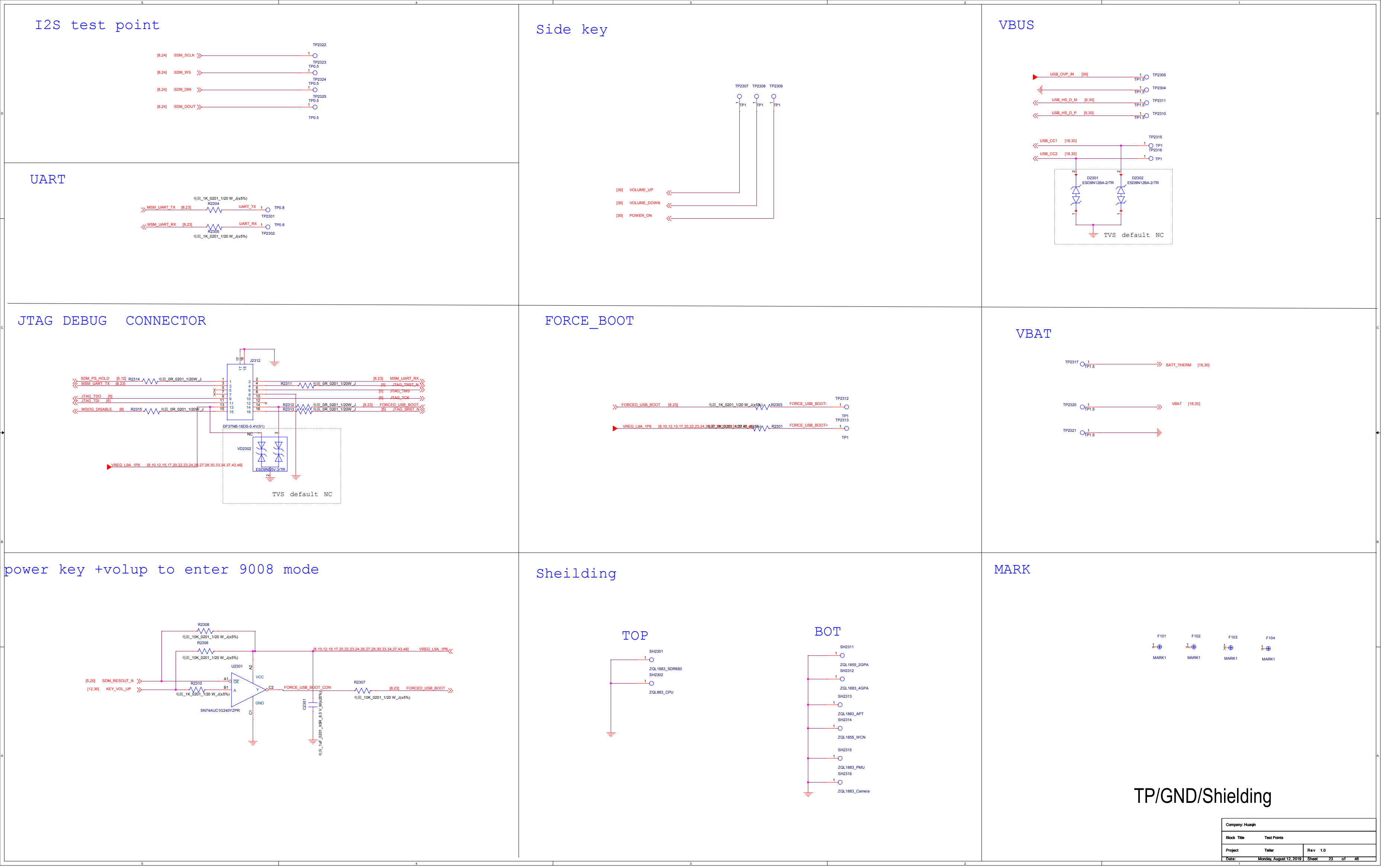
REF 2201-2299



C2207 Qualcomm suggest 10V

C2212/C2213/C2214/C2225 Qualcomm suggest resolution is K

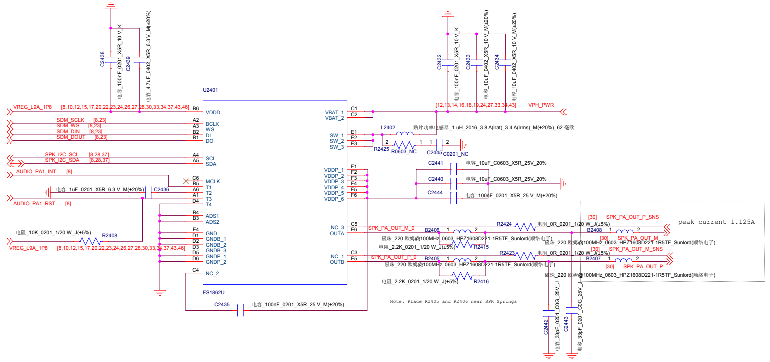
Company: Huaqin			
Block Title	Wipower		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	22 of 46



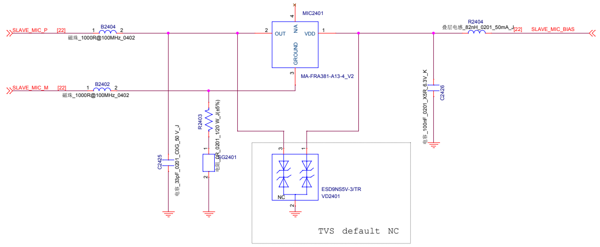
I2S test point
SDM_SCLK
SDM_WS
SDM_DIN
SDM_DOUT
TP2322
TP2323
TP2324
TP2325
TP0.5
Side key
TP2307
TP2308
TP2309
TP1
VOLUME_UP
VOLUME_DOWN
POWER_ON
VBUS
USB_OVP_IN
USB_HS_D_M
USB_HS_D_P
USB_CC1
USB_CC2
TP2305
TP2304
TP2311
TP2310
TP2315
TP2316
D2301
D2302
ESD9N12BA-2/TR
TVS default NC
UART
MSM_UART_TX
MSM_UART_RX
UART_TX
UART_RX
R2304
R2305
TP2301
TP2302
TP0.8
JTAG DEBUG CONNECTOR
J2312
SDM_PS_HOLD
MSM_UART_TX
JTAG_TDO
JTAG_TDI
WDOG_DISABLE
R2314
R2311
R2312
R2313
R2315
MSM_UART_RX
JTAG_TRST_N
JTAG_TMS
JTAG_TCK
FORCED_USB_BOOT
JTAG_SRST_N
DF37NB-16DS-0.4V(51)
VD2302
ESD9N5V-3/TR
VREG_L9A_1P8
TVS default NC
FORCE_BOOT
FORCED_USB_BOOT
VREG_L9A_1P8
R2303
R2301
FORCE_USB_BOOT-
FORCE_USB_BOOT+
TP2312
TP2313
TP1
VBAT
TP2317
TP2320
TP2321
BATT_THERM
VBAT
TP1.5
power key +volup to enter 9008 mode
R2308
R2306
R2310
R2307
U2301
SN74AUC1G240YZPR
SDM_RESOUT_N
KEY_VOL_UP
FORCE_USB_BOOT_CON
FORCED_USB_BOOT
VREG_L9A_1P8
C2301
VCC
GND
Sheilding
TOP
BOT
SH2301
ZQL1883_SDR660
SH2302
ZQL883_CPU
SH2311
ZQL1855_2GPA
SH2312
ZQL1883_4GPA
SH2313
ZQL1883_APT
SH2314
ZQL1855_WCN
SH2315
ZQL1883_PMU
SH2316
ZQL1883_Camera
MARK
F101
F102
F103
F104
MARK1
TP/GND/Shielding
Company: Huaqin
Block Title
Test Points
Project
Teller
Rev 1.0
Date:
Monday, August 12, 2019
Sheet 23 of 46

Speaker SMART PA

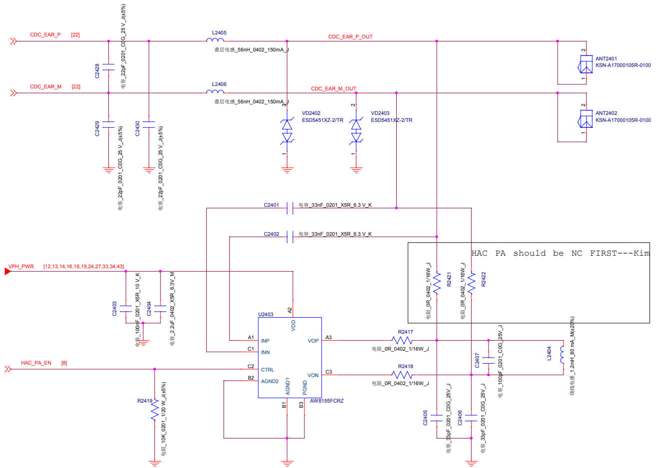
IIC Address
1nd TAS2558 0x4C
2ST FS1862 0x34



slave MIC



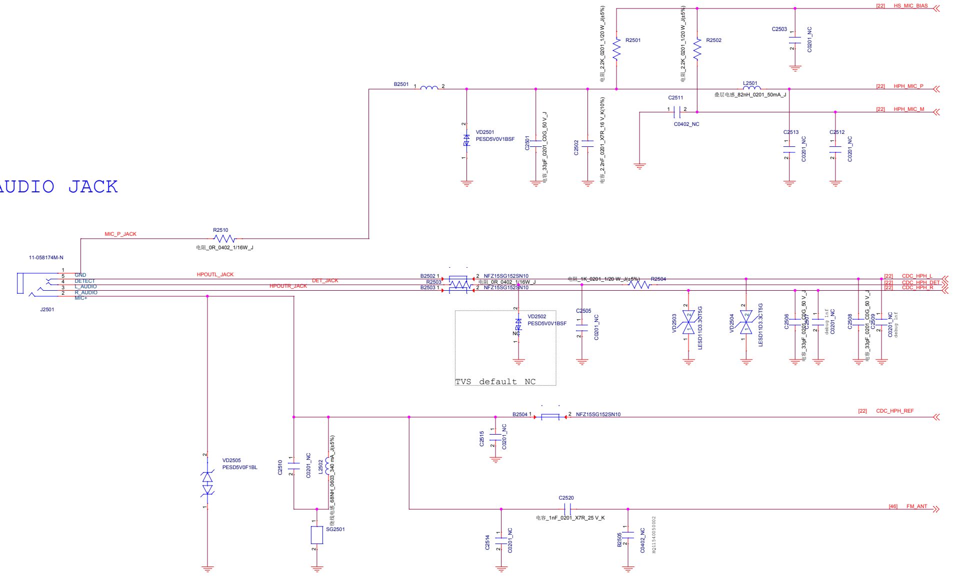
Receiver & HAC



HAC BOM list			
	HAC NC	HAC	
		PA NC	PA
C2401	NC	Y	Y
C2402	NC	Y	Y
C2403	NC	NC	Y
C2404	NC	NC	Y
C2405	NC	NC	Y
C2406	NC	Y	Y
C2407	NC	Y	Y
R2417	NC	NC	Y
R2418	NC	NC	Y
R2419	NC	NC	Y
R2421	NC	Y	NC
R2422	NC	Y	NC
L2404	NC	Y	Y
U2403	NC	N	Y

Company: Huaqin			
Block Title	AUDIO_MIC_RCV_SPK		
Project	Teller	Rev 1.0	
Date:	Saturday, September 07, 2019	Sheet 24	of 46

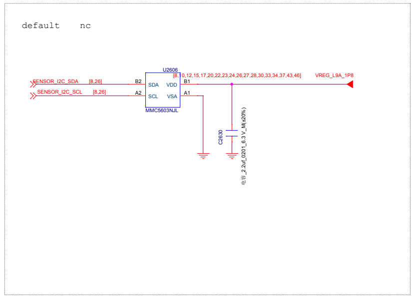
AUDIO JACK

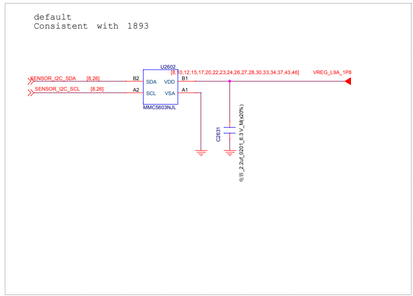


Company: Huaqin			
Block Title	AUDIO_HEADSET		
Project	Teller	Rev	1.0
Date:	Thursday, September 05, 2019	Sheet 25	of 46

Magnetic Sensor

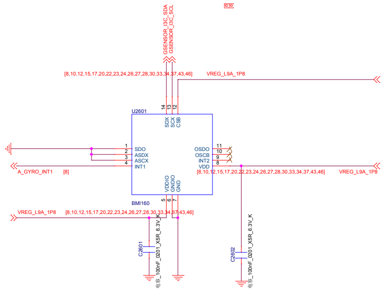
1st AK09918
2nd MMC5603NJL (Outline bigger)





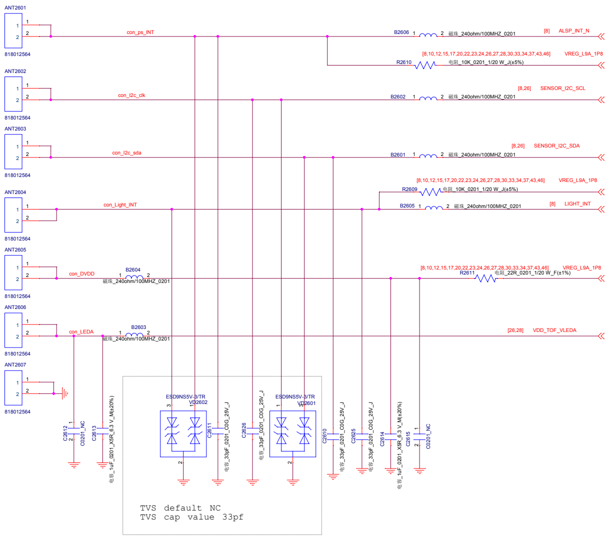
GROY Sensor

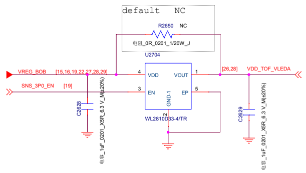
1st ST LSM6DSOW
2nd InvenSense ICM-42605



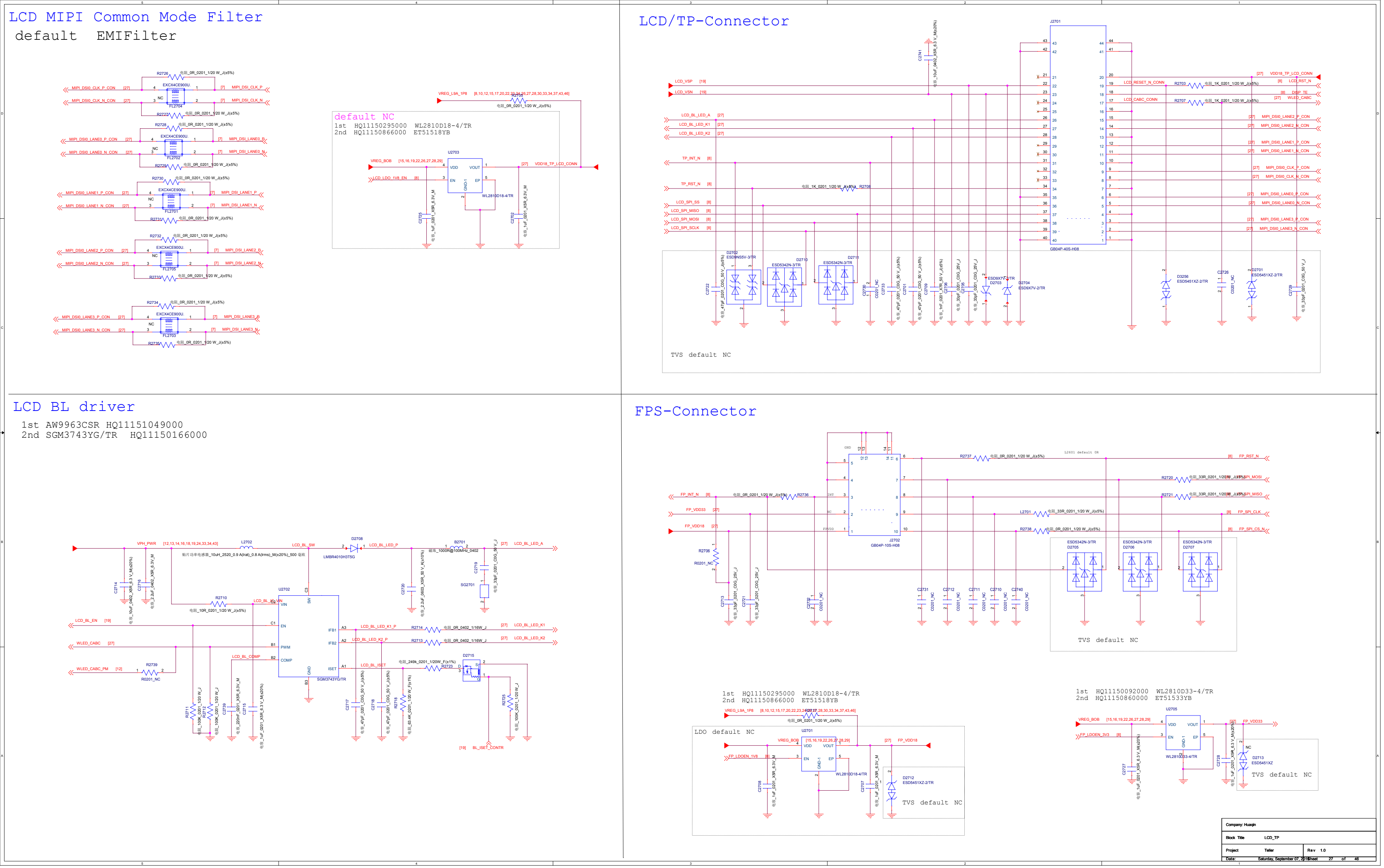
PS&ALS Sensor

Psensor:
1st LTR-2568ALS-WA
ALS:
1st TSL2540
2nd MN29005



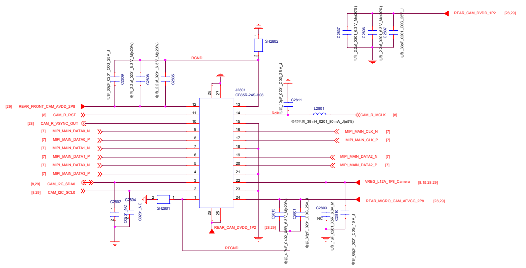


Company: Huaqin	
Block Title	SENSORS_MOTOR
Project	Teller
Rev	1.0
Date:	Saturday, July 27, 2019
Sheet	26 of 46

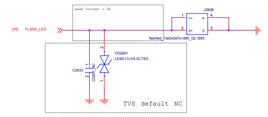

LCD MIPI Common Mode Filter
default EMIFilter
default NC
1st HQ11150295000 WL2810D18-4/TR
2nd HQ11150866000 ET51518YB
LCD/TP-Connector
TVS default NC
LCD BL driver
1st AW9963CSR HQ11151049000
2nd SGM3743YG/TR HQ11150166000
FPS-Connector
TVS default NC
1st HQ11150295000 WL2810D18-4/TR
2nd HQ11150866000 ET51518YB
LDO default NC
TVS default NC
1st HQ11150092000 WL2810D33-4/TR
2nd HQ11150860000 ET51533YB
TVS default NC
Company: Huaqin
Block Title LCD_TP
Project Teller
Rev 1.0

REAR_CAM 16M

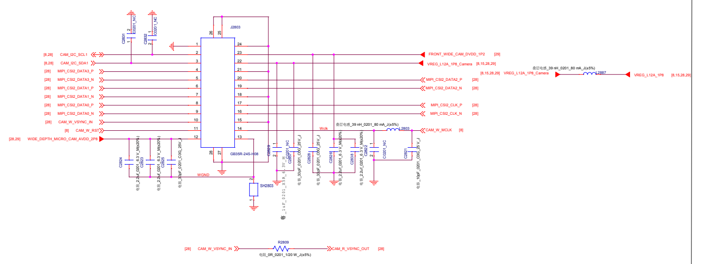
GB35A-34S-H06-E10000



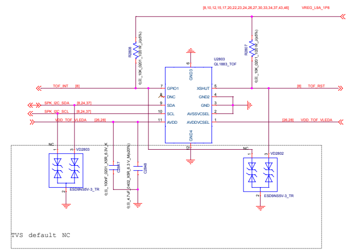
Rear Cam FLASH LED-Single CCT



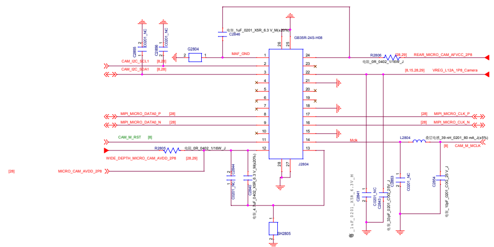
WIDE CAMERA 8M



TOF

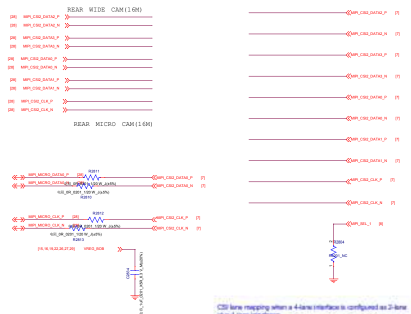


MICRO CAMERA 2M



8M&2M MIPI SWITCH

1st HQ11140405000 WAS4644C-36/TR
2nd HQ11170414000 FSA644BUCX



MIPI SWITCH		
	WIDE CAMERA (8M)	MICRO CAMERA(2M)
SEL	0	1

CSI lane mapping when a 4-lane interface is configured as 2-lane plus 1-lane interfaces:

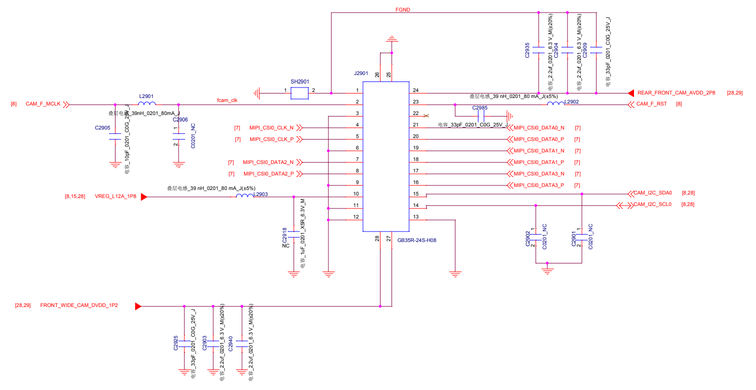
- DCLK <=> DCLK_A
- DLN0 <=> DLN0_A
- DLN1 <=> DLN1_A
- DLN2 <=> DLN0_B
- DLN3 <=> DCLK_B

CSI lane mapping when a 4-lane interface is configured as 1-lane plus 1-lane interfaces:

- DCLK <=> DCLK_A
- DLN0 <=> DLN0_A
- DLN2 <=> DLN0_B
- DLN3 <=> DCLK_B

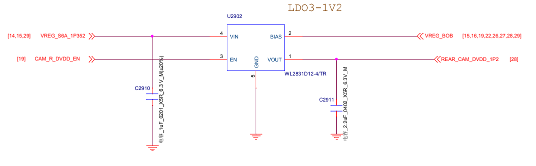
Company: Huaqin	
Block Title	REAR_CAM/FLASH
Project Teller	Rev 1.0
Date: Thursday, September 05, 2019	Sheet 28 of 46

FRONT CAMERA



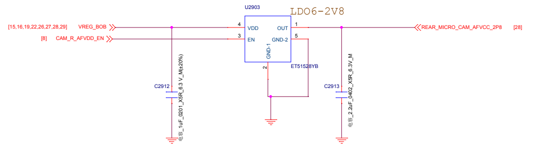
Rear DVDD 1.2V

1st HQ11150936000 WL2831D12-4/TR
2nd HQ11151048000 SGM2038-1.2XUDY4G/TR



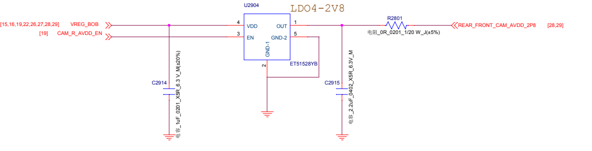
Rear&MICRO AFVCC 2.8V

1st HQ11150859000 ET51528YB
2nd HQ11150979000 WL2820X28-4/TR



Rear & Front AVDD 2.8V

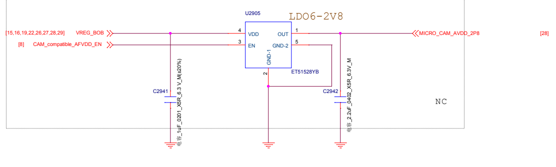
1st HQ11150859000 ET51528YB
2nd HQ11150979000 WL2820X28-4/TR



MICRO_CAM_AVDD_2P8

1st HQ11150859000 ET51528YB
2nd HQ11150979000 WL2820X28-4/TR

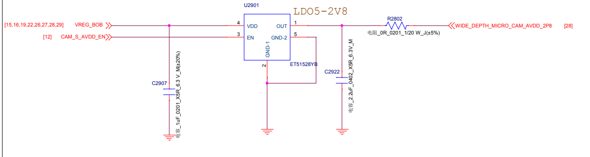
default NC



NC

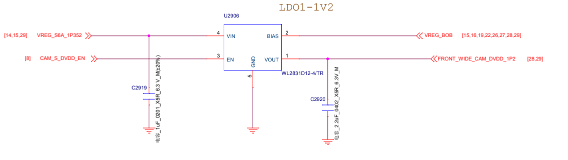
WIDE & DEPTH & MICRO AVDD 2.8V

1st HQ11150859000 ET51528YB
2nd HQ11150979000 WL2820X28-4/TR



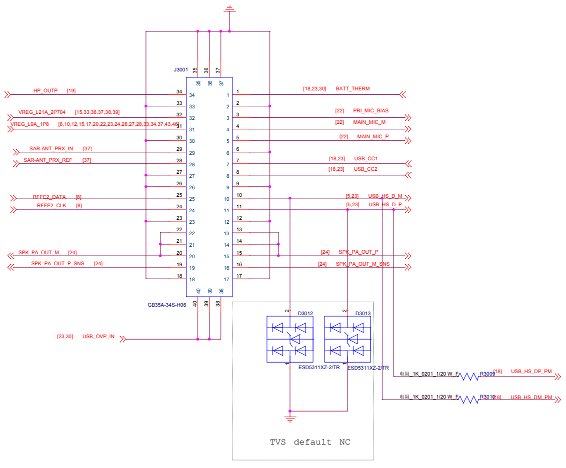
Front & Wide & Video DVDD 1.2V

1st HQ11150936000 WL2831D12-4/TR
2nd HQ11151048000 SGM2038-1.2XUDY4G/TR

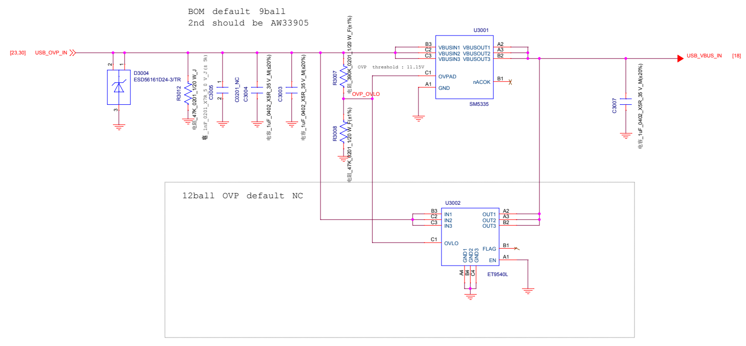


Company: Huaqin	
Block Title	FRONT_CAM/CAM_POWER
Project Teller	Rev 1.0
Date: Thursday, September 05, 2019	Sheet 29 of 46

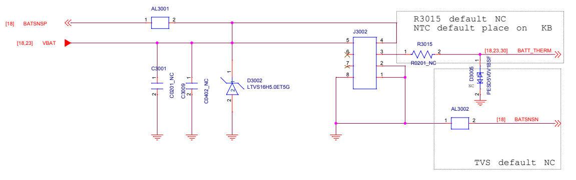
KEY_LINK_BATT



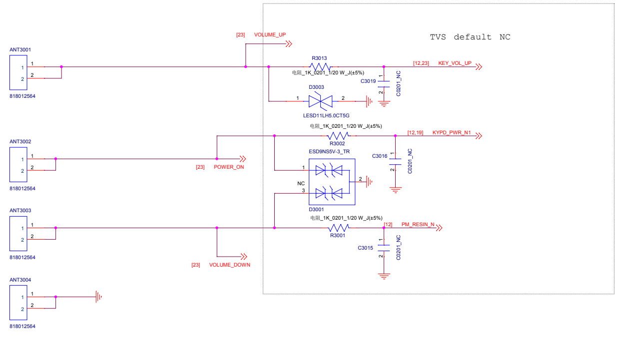
OVP



Battery Connector



Side KEY



Company: Huaqin			
Block Title	SIM_SD_KEY_LINK_BATT		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	30 of 46

DTV

REF 3101-3199

Company: Huaqin	
Block Title	DTV
Project Teller	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 31 of 46

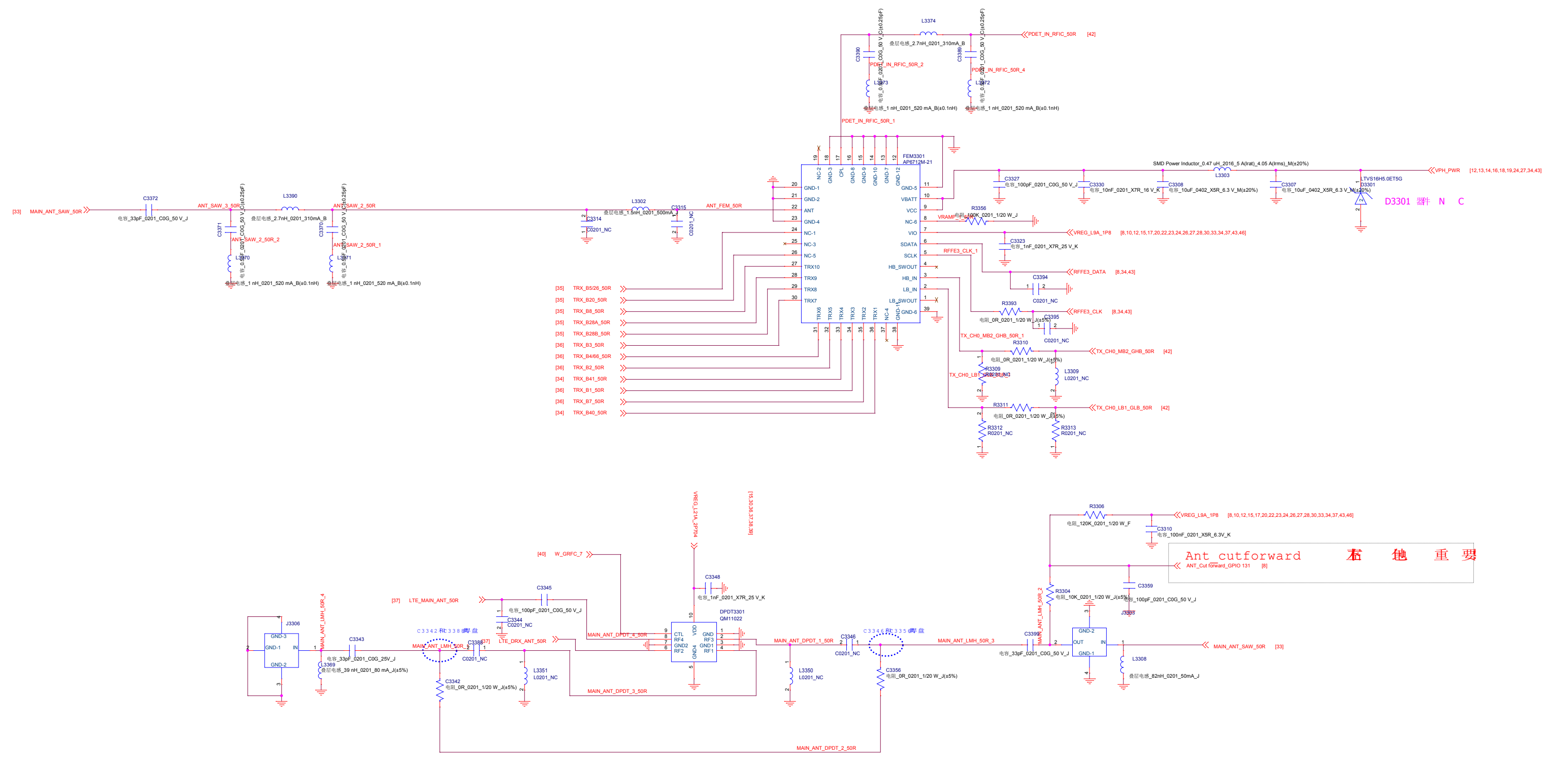
NFC

REF 3201-3299

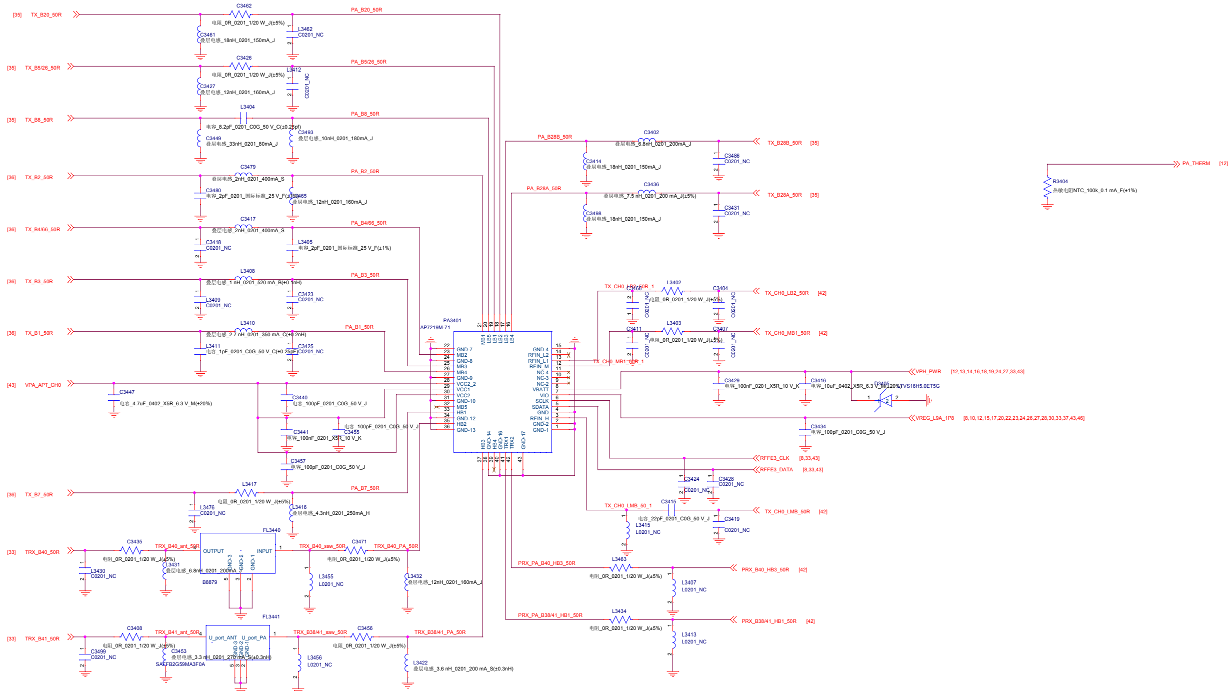
PN553   I2C Adress:0x28   (Write:0x50   Read:0x51)

Company: Huaqin	
Block Title	NFC
Project Teller	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 32 of 46

PRIMARY_ANTENNA



Company: Huaqin			
Block Title	PRIMARY_ANTENNA		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	33 of 52



QUALCOMM Technologies,Inc.
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

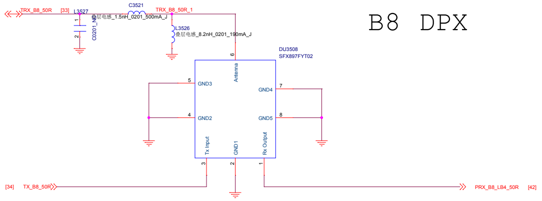
Title
SCHEMATIC, CCA, MAIN, QRD8953_2-4-2, DVT1.1

Size	Sheet Name	Rev
D	LD20-P5912	B

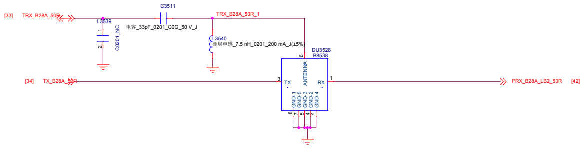
Date: Saturday, July 27, 2019

TRX_LB

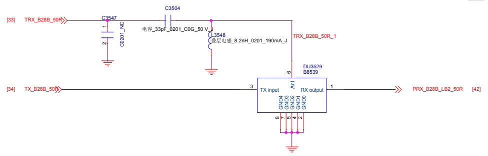
B8 DPX



B28A_PRX

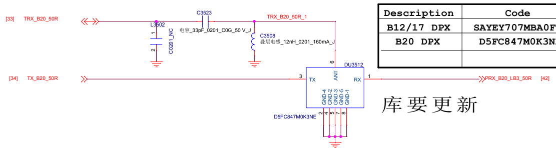


B28B_PRX



B12/17/20 DPX

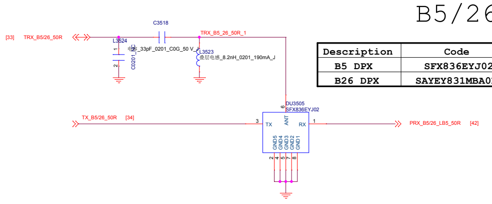
Description	Code	Vendor	SKU
B12/17 DPX	SAYEY707MBA0F0A	MURATA	Brazil & Latam
B20 DPX	D5FC847M0K3NE	TAIYO	Super_Row



库要更新

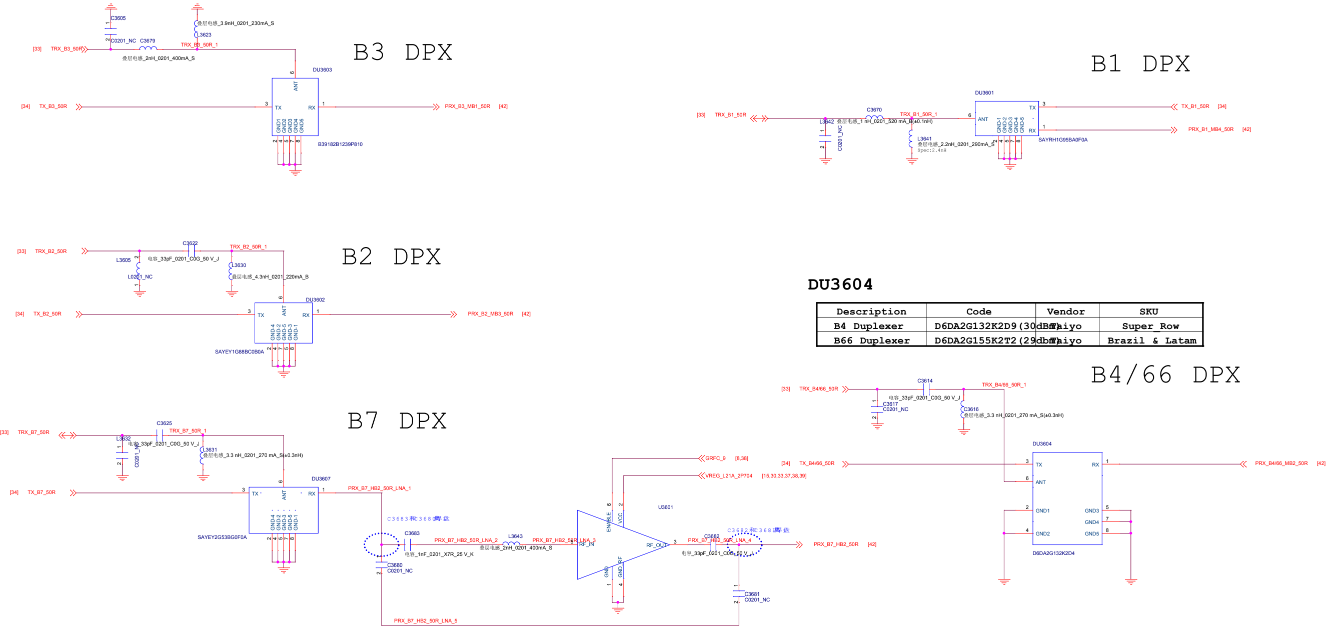
B5/26 DPX

Description	Code	Vendor	SKU
B5 DPX	SFX836EYJ02	Will	Brazil & Latam
B26 DPX	SAYEY831MBA0B0A	MURATA	Super Row



Company: Huaqin			
Block Title	TRX_LB		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	35 of 34

TRX_MHB

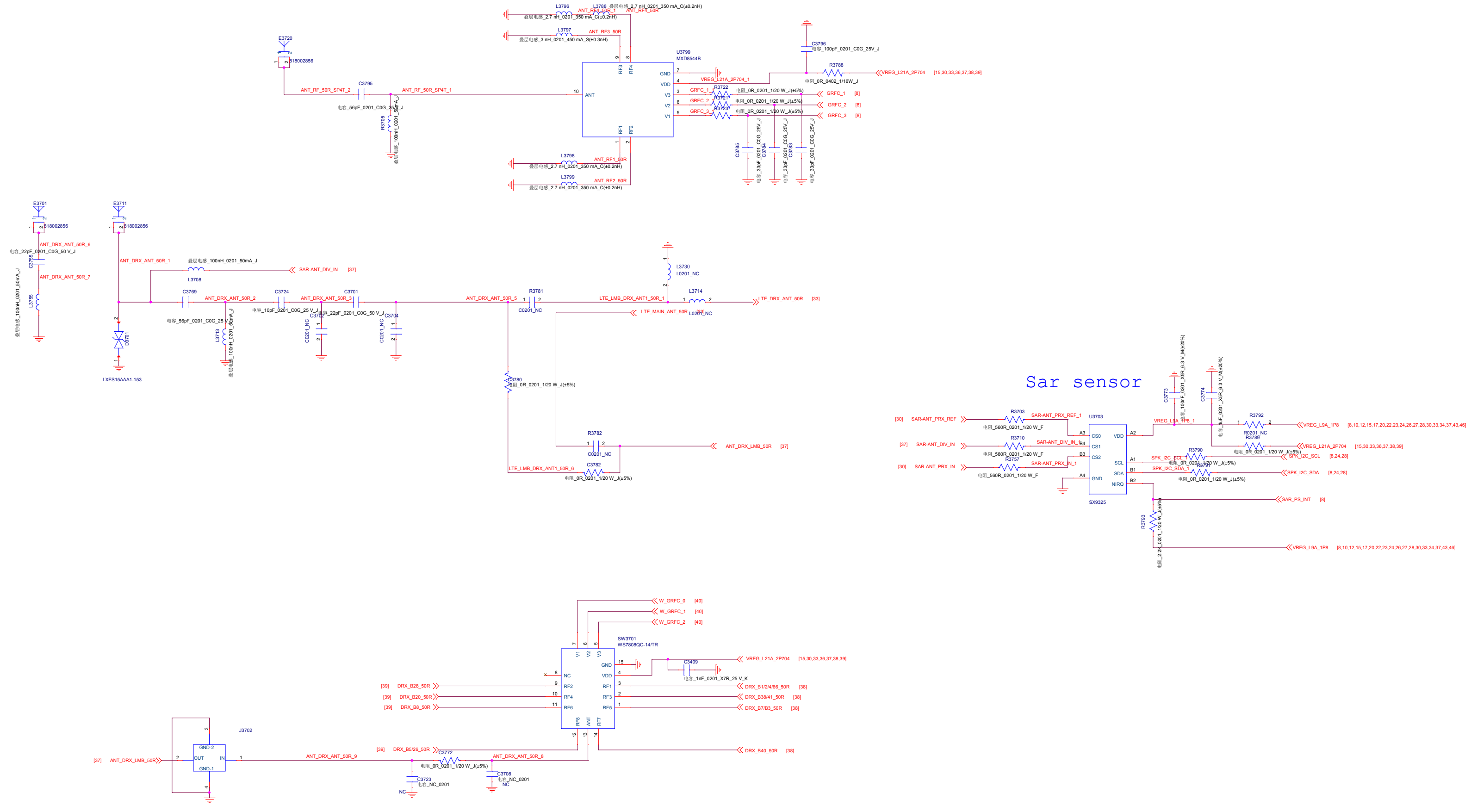


DU3604

Description	Code	Vendor	SKU
B4 Duplexer	D6DA2G132K2D9(30dBm)	Taiyo	Super_Row
B66 Duplexer	D6DA2G155K2T2(29dbm)	Taiyo	Brazil & Latam

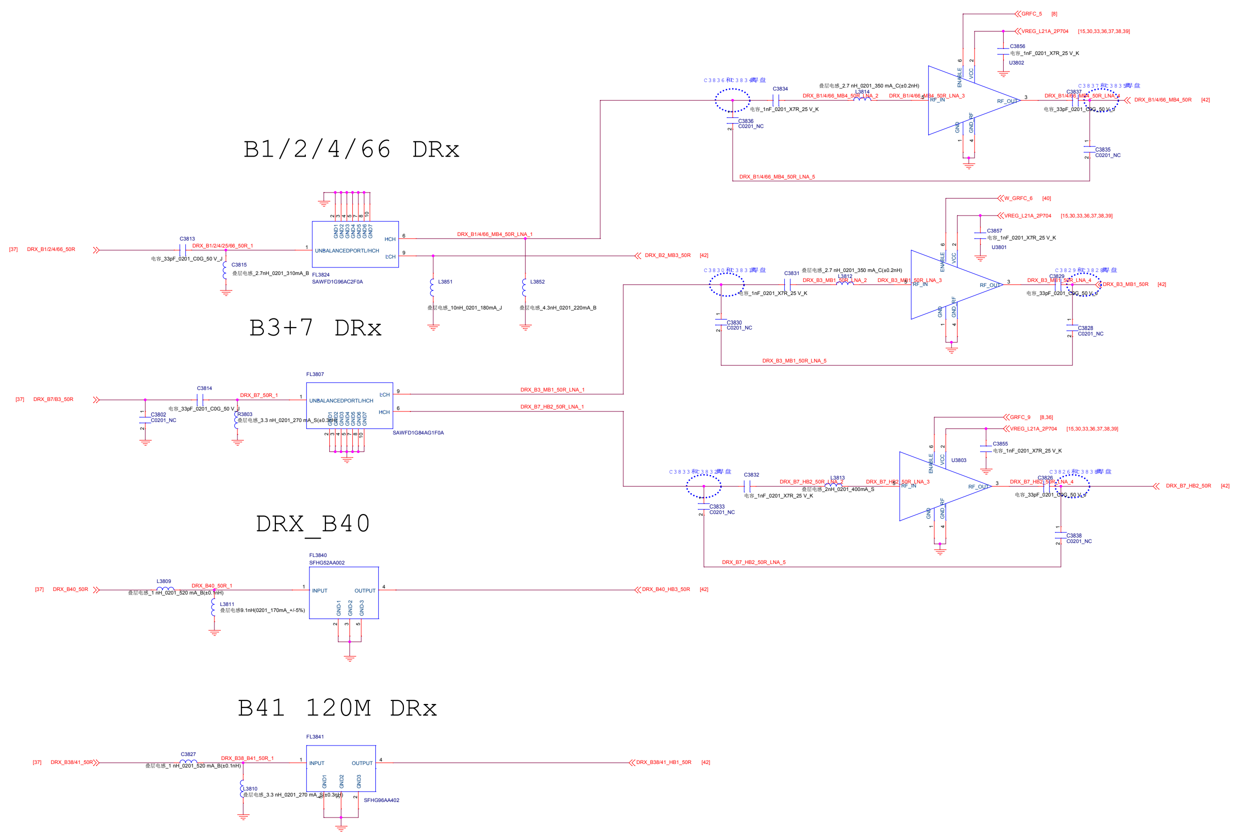
Company: Huaqin			
Block Title	TRX_MHB		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	36 of 35

DIVERSIT_ANT



Company: Huaqin			
Block Title	DIVERSIT_ANT		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet 37	of 36

DRX_MB_HB



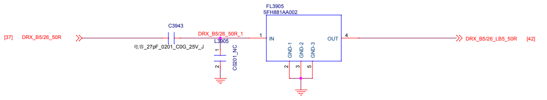
Company: Huaqin	
Block Title	DRX_MB_HB
Project Teller	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 38 of 37

NFC

DRX_LB

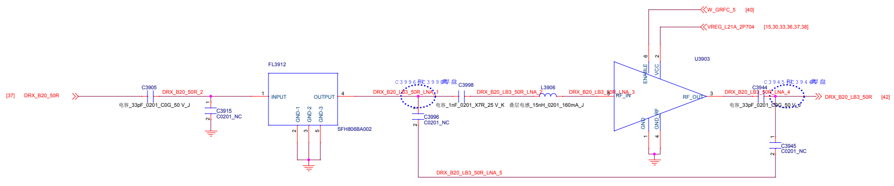
B5/26 DRX

Description	Code	Vendor	SKU
B5 SAW	SFH881AA002	Wisol	Brazil & Latam
B26 SAW	SAFFB876MAA0F0A	MURATA	Super Row

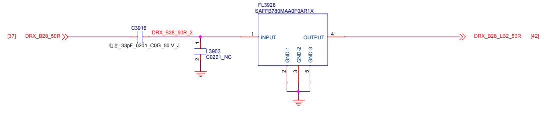


B12/17/20 DRX

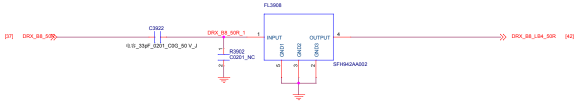
Description	Code	Vendor	SKU
B12/17 DPX	SAFFB737MAA0F0A	MURATA	Brazil & Latam
B20 DPX	SFH806BA002	Wisol	Super_Row



B28 DRX

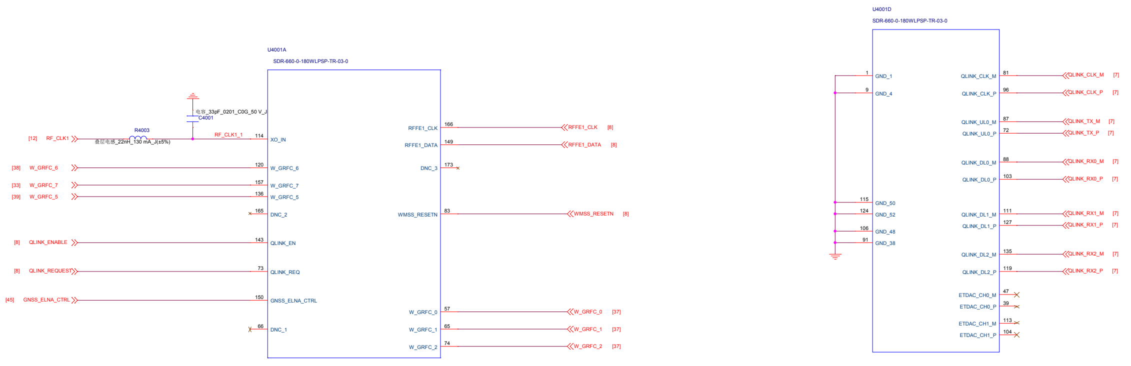


B8 DRX



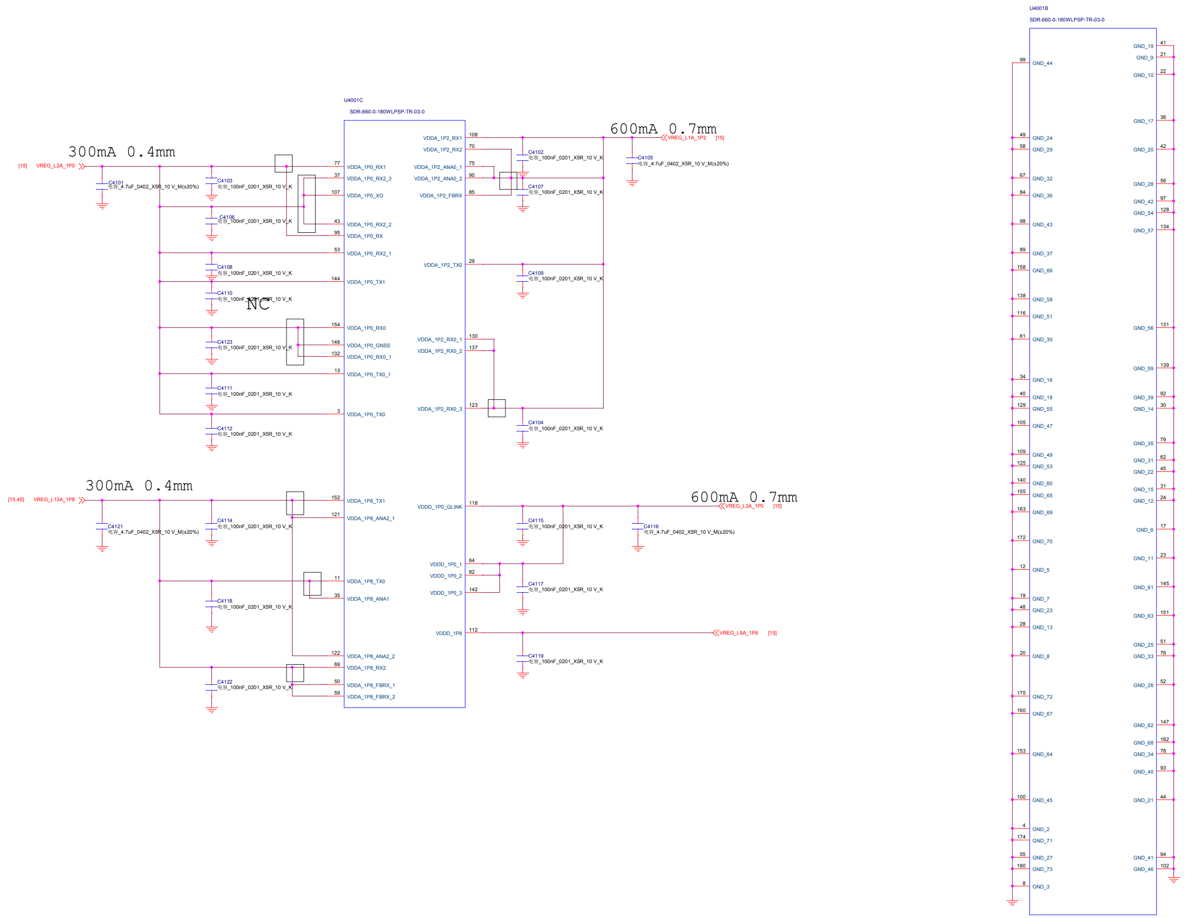
Company: Huaqin			
Block Title	DRX_LB		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	39 of 38

SDR660_MSS



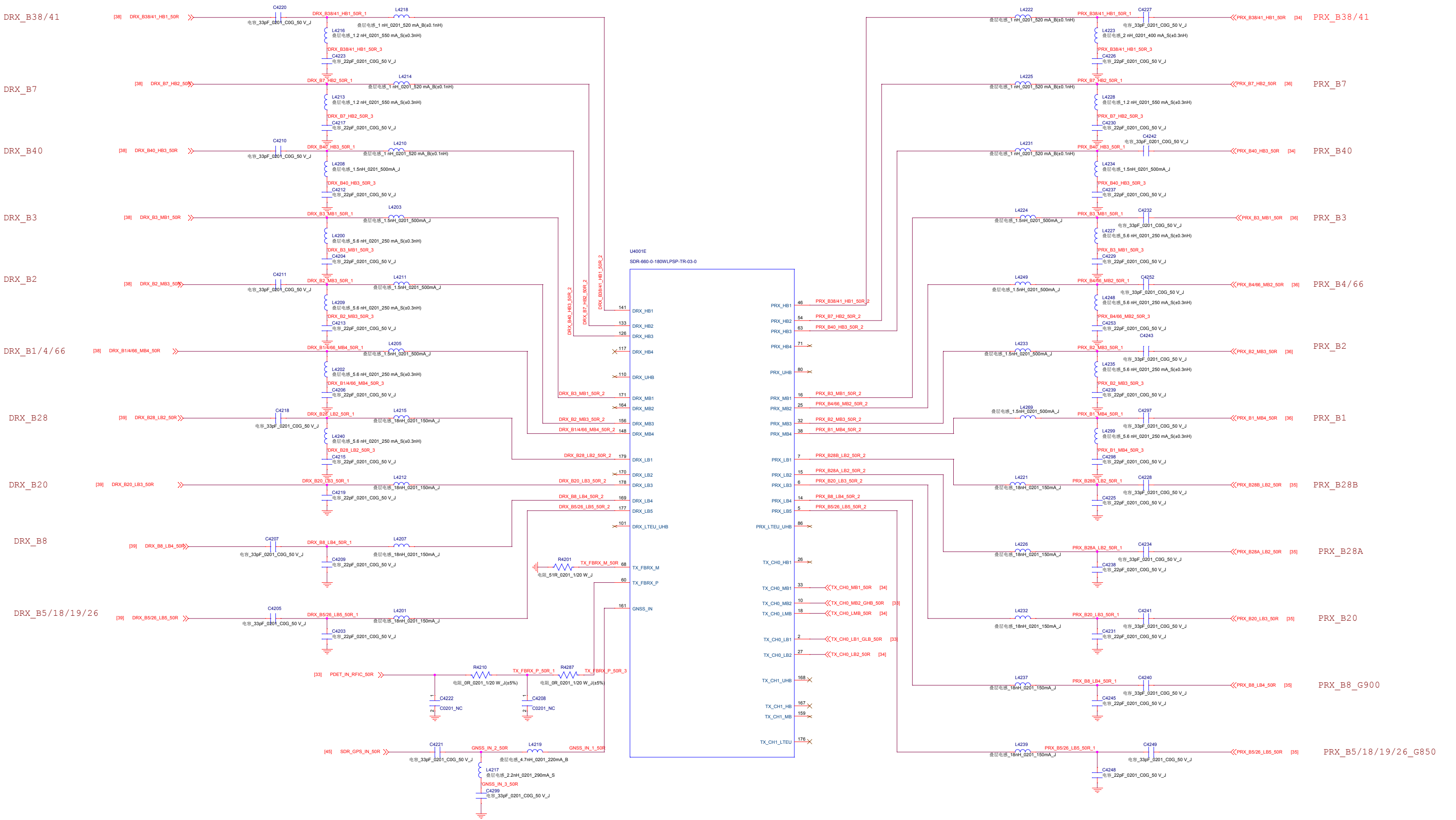
Company: Huaqin			
Block Title	SDR660_MSS		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	40 of 46

SDR660_PWR



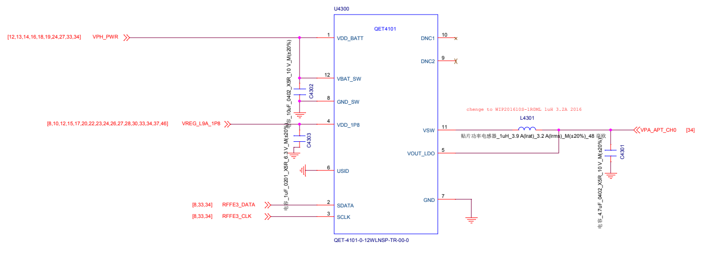
Company: Huaqin			
Block Title	SDR660_PWR		
Project	Teller	Rev	1.0
Date:	Monday, July 29, 2019	Sheet	41 of 46

SDR660_DA/PRX/DRX/FBRX



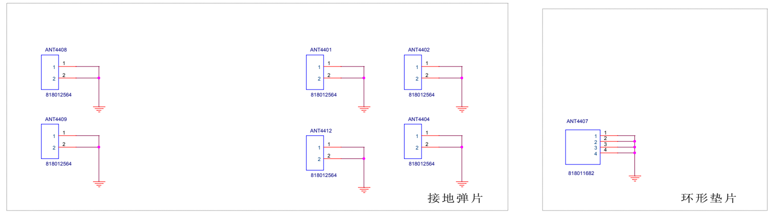
Company: Huaqin			
Block Title	SDR660_DA/PRX/DRX/FBRX		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	42 of 46

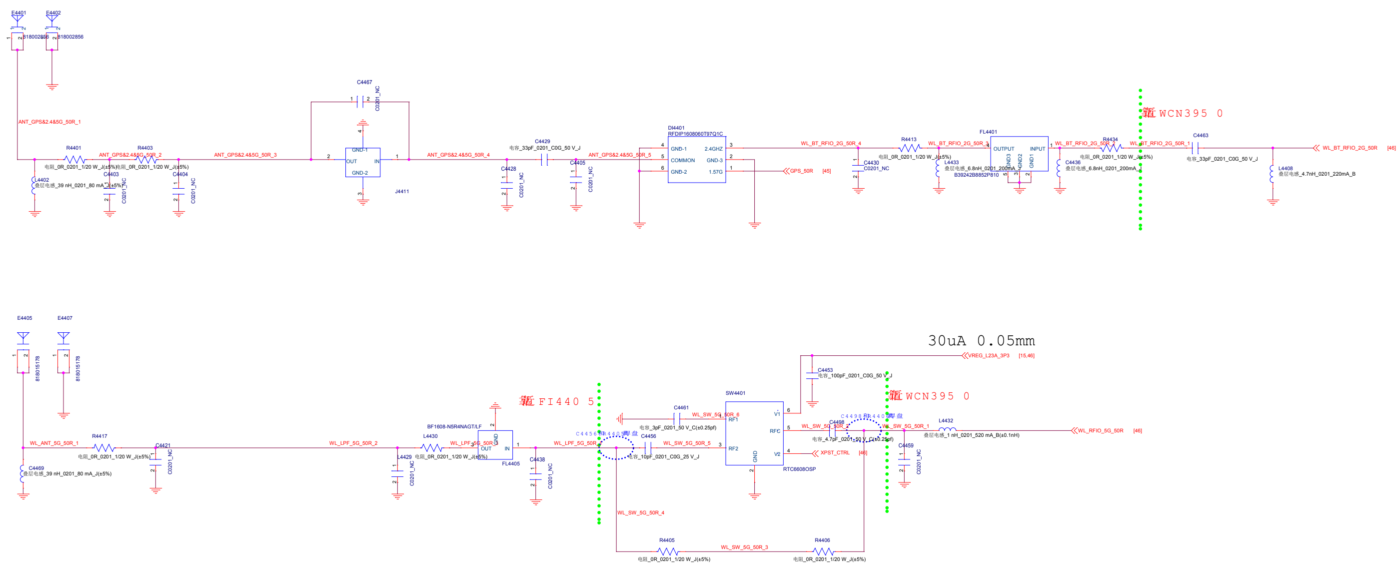
QET4101



Company: Huaqin	
Block Title	QET4101
Project Teller	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 43 of 46

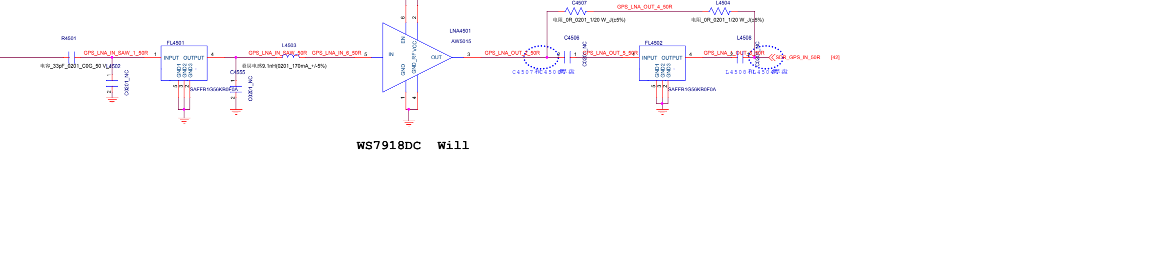
WIFI 2.4G/BT





Company: Huaqin			
Block Title	WIFI 2.4G/5G/BT		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	44 of 46

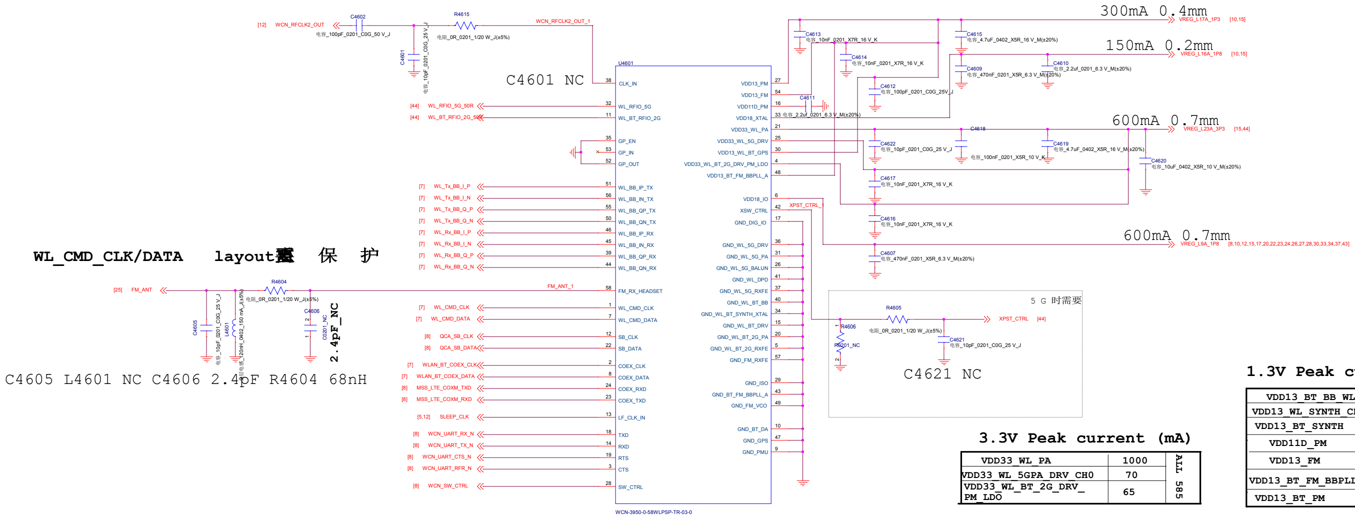
GPS



WS7918DC Will

Company: Huaqin			
Block Title	GPS		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	45 of 46

WCN_3950



3.3V Peak current (mA)

VDD33_WL_PA	1000	ALL 585
VDD33_WL_5GPA_DRV_CH0	70	
VDD33_WL_BT_2G_DRV_ PM_LDO	65	

1.3V Peak current (mA)

VDD13_BT_BB_WL	120	ALL 240
VDD13_WL_SYNTH_CH0	40	
VDD13_BT_SYNTH	15	
VDD11D_PM	20	
VDD13_FM	15	
VDD13_BT_FM_BBPLL	10	
VDD13_BT_PM	20	

1.8V Peak current (mA)

VDD18_IO	15	ALL75
VDD18_XTAL	60	

Company: Huaqin			
Block Title	WCN3980		
Project	Teller	Rev	1.0
Date:	Saturday, July 27, 2019	Sheet	46 of 46

DTV

REF 3101-3199

Company: Huaqin	
Block Title	DTV
Project Teller	Rev 1.0
Date: Saturday, July 27, 2019	Sheet 47 of 46